

DSP TO JSFX

THE USER MANUAL

79 plugins for REPEER: instruments, effects, MIDI tools, meters and more, all running as JSFX.

BY CHMAHA / FOR REAPER

WELCOME

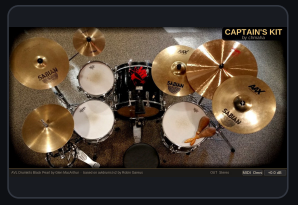
What's on the bench.

Open-source DSP brought to REAPER as JSFX, with custom interfaces. Every page here is a quick start written around what you see and click on screen. Click any plugin to jump to it.

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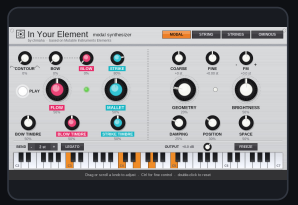
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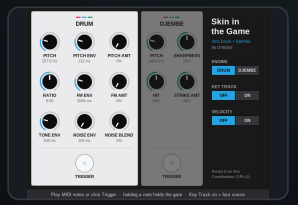
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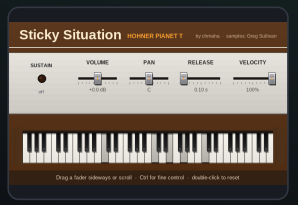
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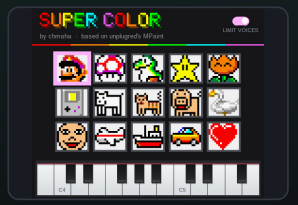
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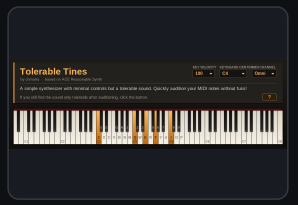
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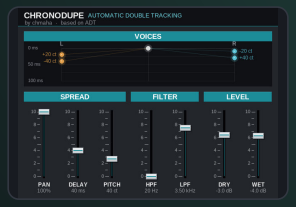
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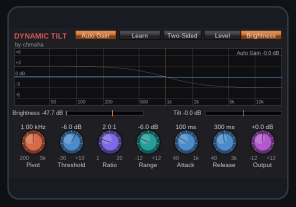
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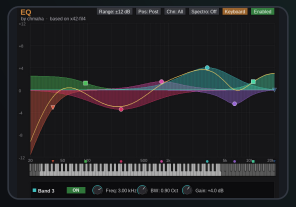
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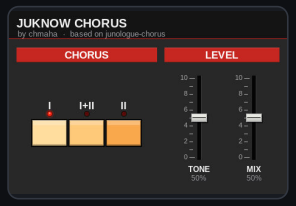
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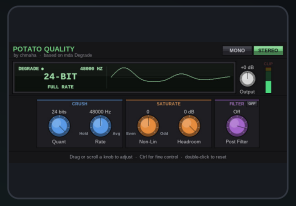
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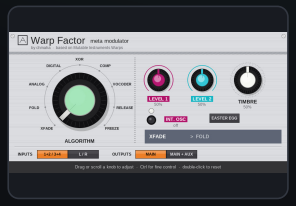
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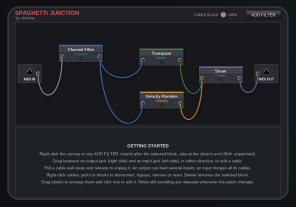
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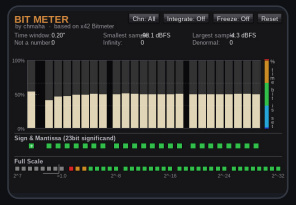


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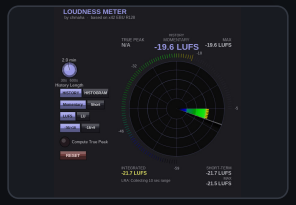
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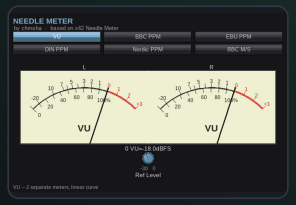
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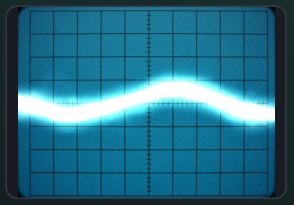
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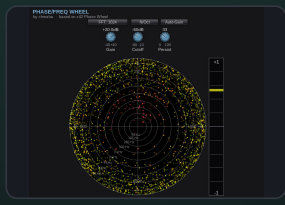
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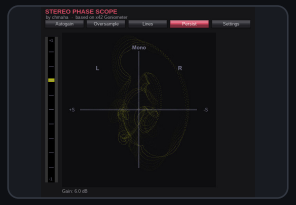
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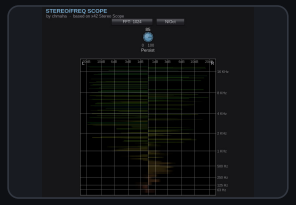
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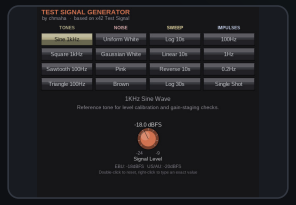
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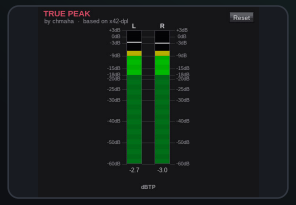
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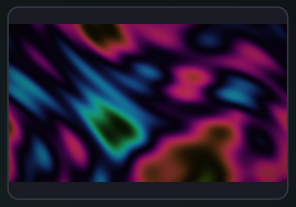


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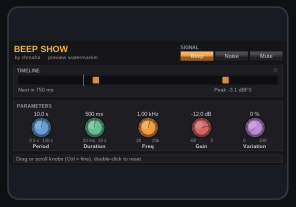
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GETTING STARTED

Plug in.

Install with ReaPack

1. In REAPER, open **Extensions > ReaPack > Import repositories** and paste this address:

   `https://github.com/chmaha/DSP-Ports-JSFX/raw/main/index.xml`

2. Open **Extensions > ReaPack > Browse packages**, filter by **DSP to JSFX** and install the plugins you want.
3. The plugins then appear in the FX browser under **JS**.

ReaPack also installs each plugin's **Dependencies** files and keeps everything up to date.

Or install by hand

Copy the plugin's **.jsfx** file into your REAPER **Effects** folder (**Options > Show REAPER resource path**). If the plugin has files in **Dependencies** (the drum kits, the unplugred ports and others), also copy the **Dependencies** folder that sits next to the .jsfx, and keep it next to the .jsfx. Each plugin needs its own subfolder, such as Drumkits, Scope or VU.

IN EVERY PLUGIN

Knobs

On the effects with round knobs, drag a knob up or down, use the mouse wheel to nudge it, and double-click to reset it to its default. Many plugins also offer fine adjustment: hold Ctrl or Shift while dragging. Each page tells you where that applies.

Automation

Everything is automatable from REAPER too. The names on these pages are the labels you see in each plugin's own interface.

Looking for Break Room?

Fifteen mini games for when the mix needs you to walk away from it. It lives in this collection too, and has its own manual:

https://chmaha.github.io/DSP-Ports-JSFX/BreakRoom-Manual.pdf



INSTRUMENTS

Synths, samplers, organs and drum machines. Put one on a track, send it MIDI and play.

24 ENTRIES



INSTRUMENTS / 1 OF 24

Acid Reign

A TB-303-style acid bass with the Devil Fish mods and a 16-pattern sequencer.

Based on Open303

A monophonic acid bass synth in the style of the TB-303. The top row has **TUNING** and **WAVEFORM** (a blend from sawtooth to square), the **FILTER** knobs (**CUTOFF**, **RESONANCE**, **ENV MOD**, **DECAY**), **ACCENT** and **VOLUME**. **DEVIL FISH** adds **SLIDE** time, filter **ATTACK**, **ACC DECAY**, **AMP DECAY**, **RELEASE** and **SUSTAIN**. **FILTER MODE** picks one of 16 filter responses from the menu, with quick buttons for **LP 18** (the default) and **TB 303**.

From MIDI, notes with velocity 100 or more are accented, and a note played while another is held slides to it. CC 74, 71, 81 and 7 control cutoff, resonance, env mod and volume; pitch bend covers ±12 semitones.

The **SEQUENCER** buttons choose **OFF** (play from MIDI), **KEY SYNC** (the pattern runs while you hold a key, transposed by that key) or **HOST SYNC** (the pattern follows REAPER's transport and tempo, starting from the **ROOT** note unless a key is held; drag or scroll ROOT, double-click for C2). Pick one of 16 patterns with the numbered buttons; while the sequencer runs, the new pattern starts when the current one ends (its button blinks until then). In the grid, click a cell to set a step's note and click the lit cell again for a rest. The rows below set each step's **OCT**ave (click or wheel up, right-click or wheel down), **ACC**ent and **SLIDE** into the next step. **<** and **>** shift the pattern, **RANDOM** rolls a new one, **CLEAR** empties it and **GATE** sets how long each step's note lasts. Patterns are saved with the project.

AVL Drumkits

Captain's Kit, Led Head, Goldibops and Busk Stop. Click the photo to hear each piece.

Based on avldrums.lv2, with samples from Glen MacArthur's AVL Drumkits



Sampled kits (Black Pearl, Red Zeppelin, Blonde Bop and Buskman's Holiday). **Click a piece of the kit photo** to hear it, and use the **mouse wheel** over the photo to change the audition velocity. The pieces light up when played from MIDI on the General MIDI drum notes (36–61; Goldibops and Busk Stop start at 35). Hi-hats and cymbals choke.

The controls are **Level**, **MIDI Channel** and **Display** (text, highlight or both). Goldibops has a **Sticks / Hot Rods** switch. Captain's Kit, Led Head and Goldibops can also output separate channels (kick, snare, hi-hat, tom, floor tom, overheads, percussion); Busk Stop is stereo only.

AVL Drumkits

Led Head



Goldibops



Busk Stop





Chamber Organ

A five-stop pipe organ with 80 historical temperaments.

Based on Aeolus

Click the stops (Principal 8', Flute 8', Principal 4', Flute 4', Octave 2') to switch them on and off. Choose a **Temperament** from the list of 80 historical tunings and set the concert pitch. **Retune** applies the new tuning. **Reverb Settings** opens delay, time, width and amount. There is also a **Volume** control with an output meter and a **MIDI Channel** choice.



- **MASTER SYS**: three tone channels and noise (white or periodic, with a clock that can follow tone 3 for bass). Panning works in the Game Gear model.
- **MEGA DRIVE**: YM2612 or YM3438 FM with a full patch editor: algorithm (shown as a diagram), feedback, LFO and sensitivities, and four operator panels with rates, levels, multiple, detune, key scale, AM and SSG-EG.
- **PC ENGINE**: a 32-step wavetable (drag to draw your own, or right-click it or use **WAVE** for eight preset shapes), an envelope, noise (click the NOISE title to play it from the keyboard) and a channel display.
- **SNES**: 203 BRR samples by Doug Fraker with four drum kits. **INSTRUMENT** chooses the default sound, and the **CHANNELS** panel gives each MIDI channel its own sound (channel 10 chooses the kit); Program Change and Bank Select also pick sounds. **ENVELOPE** uses each instrument's own ADSR or a custom one, and **ECHO** has delay, feedback, volume and the 8-tap FIR filter.

A drawn wave is the **Custom** shape and is saved with the project; it is loaded when the next note starts, as on the real chips. Click or drag along the on-screen keyboard to play (lower on a key is louder). Drag knobs, hold Ctrl or Shift for fine control, use the wheel to nudge, and double-click to reset.

Console Yourself

The sound chips of six game consoles, from the NES to the Super NES.

Based on Nes_Snd_Emu, MAME, ymfm and snes_spc

The sound chips of six game consoles. Pick one with the tabs at top left: **NES**, **GAME BOY**, **MASTER SYS** (Master System or Game Gear), **MEGA DRIVE**, **PC ENGINE** or **SNES**. The display shows the current settings and the scope shows the output.

- **GLOBAL**: **MIDI MODE Layer** plays every note on the same set of channels (**LAYER** chooses which; on the NES, Game Boy and Master System it can also be a single channel, such as the triangle alone for bass), and **CHIPS** stacks up to 16 copies of the chip so chords can be played; MIDI channel 10 plays drums. **Tracker** maps MIDI channels to the chip's hardware channels, as game music drivers did. Also **BEND** range, **VELOCITY** on or off, **GLIDE**, **ARPEGGIO** speed, **VIB DELAY**, **KEYS** (which channel the on-screen keyboard plays; clicking a channel's title does the same) and **VOLUME**. The **SUSTAIN** lamp lights while the pedal is down.
- **NES**: two pulse channels (duty, envelope, vibrato, detune and hardware sweep), triangle, and noise plus DPCM drums. **OUTPUT** chooses the Famicom, NES or clean output filter. **EXPANSION** adds a Konami **VRC6** (two pulse channels with eight duties and a sawtooth) or **VRC7** (six FM channels with 15 built-in patches and one custom patch); the **2A03** and **VRC6**/**VRC7** tabs switch between the two pages, and **EXPANSION LAYER** sets how many expansion channels join each note.
- **GAME BOY**: two pulse channels with the hardware volume envelope, a wave channel (drag on the wave view to draw your own wave, or right-click it for eight preset shapes), noise, and left, centre or right panning.

Console Yourself

NES + VRC7



MEGA DRIVE



SNES





Dr. Nine

A drum synthesizer with eleven voices and a 16-pattern step sequencer.

Based on 9W9

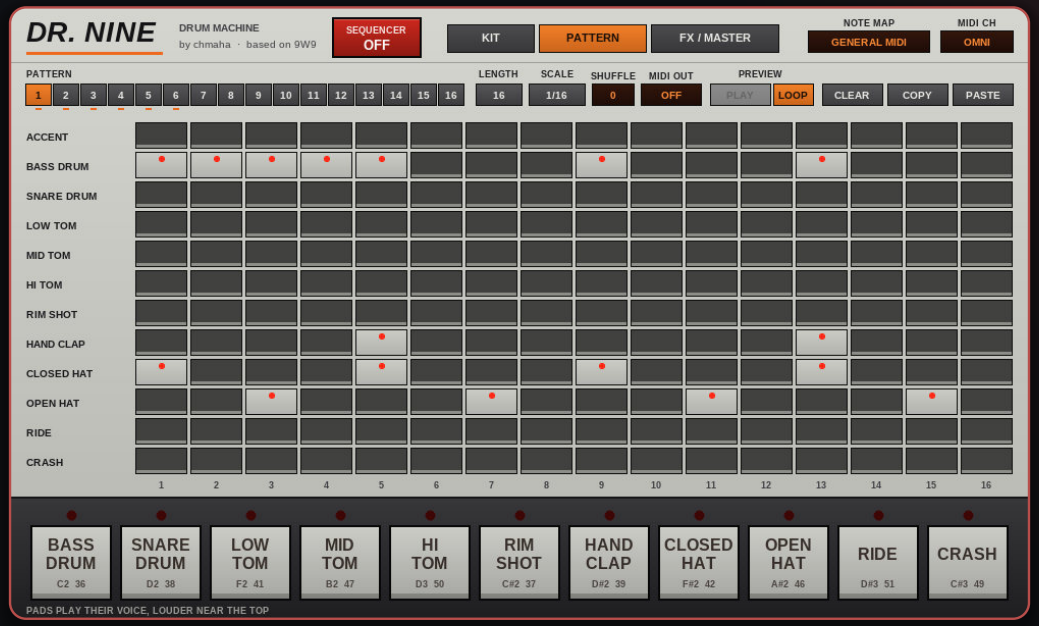
Three pages: **KIT**, **PATTERN** and **FX / MASTER**.

- **KIT**: knobs for each voice (kick, snare, three toms, rim shot, clap, hi-hats, ride, crash): tune, decay, drive, a distortion type (Diode, Clip, SAT, BFZ, PDIST, Fold, Crush), pan, level, and reverb and delay sends.
- **PATTERN**: a 16-pattern step sequencer. The big **SEQUENCER** ON/OFF button (visible on every page) arms the sequencer. When it is ON (green), the selected pattern plays and loops in sync with REAPER's transport whenever the project is playing or recording, and the button reads RUNNING with a blinking dot on the beat. When it is OFF (red), the sequencer stays silent and Dr. Nine only responds to incoming MIDI. Pattern changes made while it runs take effect when the pattern next loops. It has **PLAY/STOP** (a preview that works only while REAPER's transport is stopped, with an optional **LOOP**), **SHUFFLE**, **ACCENT**, **COPY / PASTE / CLEAR** (Shift-click copies a single row or all patterns) and **MIDI OUT** to send the pattern to other instruments.
- **FX / MASTER**: reverb, delay, master distortion and the glue compressor.

Play it from MIDI using the **NOTE MAP** (General MIDI drum notes or chromatic from C1). Drag knobs, use Ctrl or Shift for fine control, the wheel to nudge, and double-click to reset.

Dr. Nine

PATTERN



FX / MASTER



INSTRUMENTS / 6 OF 24

Evangelina Symphonica Espressivo String Ensemble

A sampled string ensemble with seven key-switched articulations.

Based on the sfizz engine, playing S. Christian Collins' SCC Expressive Strings



A sampled string ensemble. The **INSTRUMENT** page has the **CONTROLS** choice (**Performance**, with dynamics on the mod wheel, or **Notation**, with dynamics from velocity) and the **ARTICULATION** list: legato (expression or velocity), staccato, tremolo (expression or velocity), legato / tremolo crossfaded with the soft pedal (CC67) and pizzicato. Click an articulation, or play its key switch, to select it. The big dial in the middle shows the live expression value; the **Expression controller** menu picks whether it follows the mod wheel, breath, foot or CC11.

Along the bottom, **Level** and **Pan** set the part, **MIDI channel** picks Omni or one channel, the **Sustain** and **Sostenuto** lamps show the pedals, and the status text shows voices in use or loading progress. **Output** (top right) sets the master level. Click **SETTINGS** at the top left for **Concert pitch (A4)**, **Temperament** and **Presets**, **Polyphony**, **Sample quality** and **Pitch bend range**. The clickable keyboard plays the instrument. The first load reads the sample file, so allow a moment.

INSTRUMENTS / 7 OF 24

Grand Designs

A sampled Yamaha CP80 electric grand with half-pedalling.

Based on the sfizz engine, playing Greg Sullivan's CP80 samples

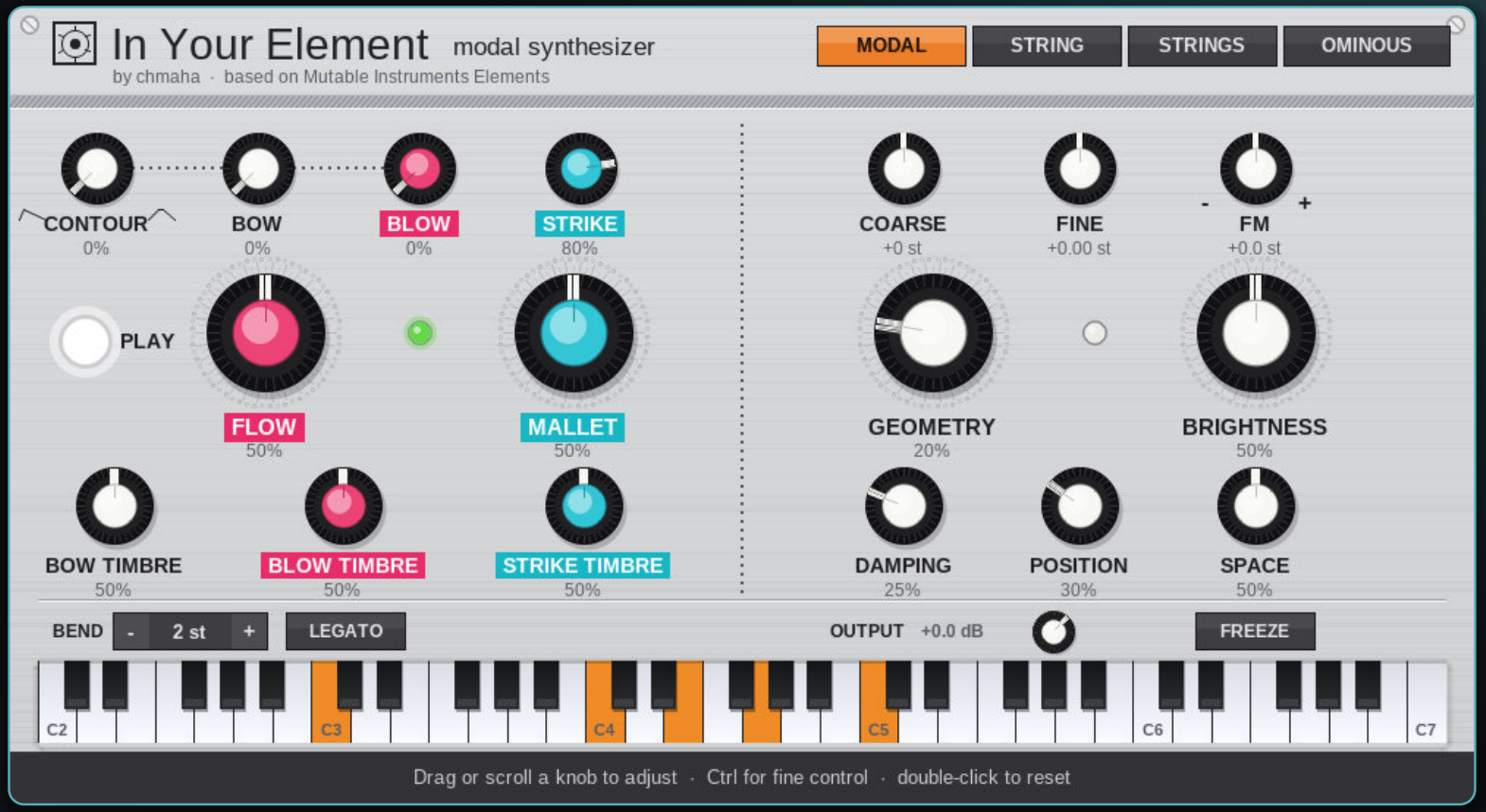


A sampled Yamaha CP80 electric grand, using Greg Sullivan's samples. Four velocity layers, with the sustain loops stored in the samples. Drag or scroll a knob (Ctrl for fine control) and double-click to reset it. **VOLUME** sets the output level (plus or minus 12 dB), **PAN** (CC10) the position in the stereo field, **RELEASE** (CC72) how long notes take to fade after the key comes up (0 to 2 seconds) and **VELOCITY** (CC99) how much key velocity changes the level (0% plays every note at full level). The **SUSTAIN** lamp glows as the dampers lift: click it to hold or release, or scroll or drag it for half-pedal. Pedalling from MIDI (CC64) works the same way, and pressing the pedal again catches notes that are still dying away. Click the keyboard to play (88 keys, A0 to C8); lower on a key is louder. Needs the Dependencies/GrandDesigns files; a bar shows the samples loading.

In Your Element

Bow, blow and strike exciters feeding a modal resonator.

Based on Mutable Instruments Elements



The left half is the exciter: **Contour** shapes the envelope of the bow and blow exciters (percussive to sustained to slow swell); **Bow**, **Blow** and **Strike** set their levels; **Flow** and **Mallet** pick the blow and strike character (the mallet goes from sampled hits through mallet and plectrum to bouncing particles); the three **Timbre** knobs set their brightness. **Play** holds the gate. The right half is the resonator: **Coarse**, **Fine** and **FM** (mod wheel pitch depth), **Geometry** (shape of the resonator, or the chord in Strings mode), **Brightness**, **Damping**, **Position** and **Space** (stereo width into reverb; **Freeze** holds the reverb). The engine buttons choose Modal, String, Strings or Ominous. Velocity sets the strength, channel pressure changes it while a note sounds, **Legato** slides between held notes and **Bend** sets the pitch-bend range. The left input excites the strike path, the right input the blow path. Click the keyboard to audition. Needs the Dependencies/InYourElement files. Drag or scroll a knob (hold Ctrl for fine adjustment) and double-click to reset it.

JamSynth

A subtractive synth with 28 factory banks and a 128-slot user bank.

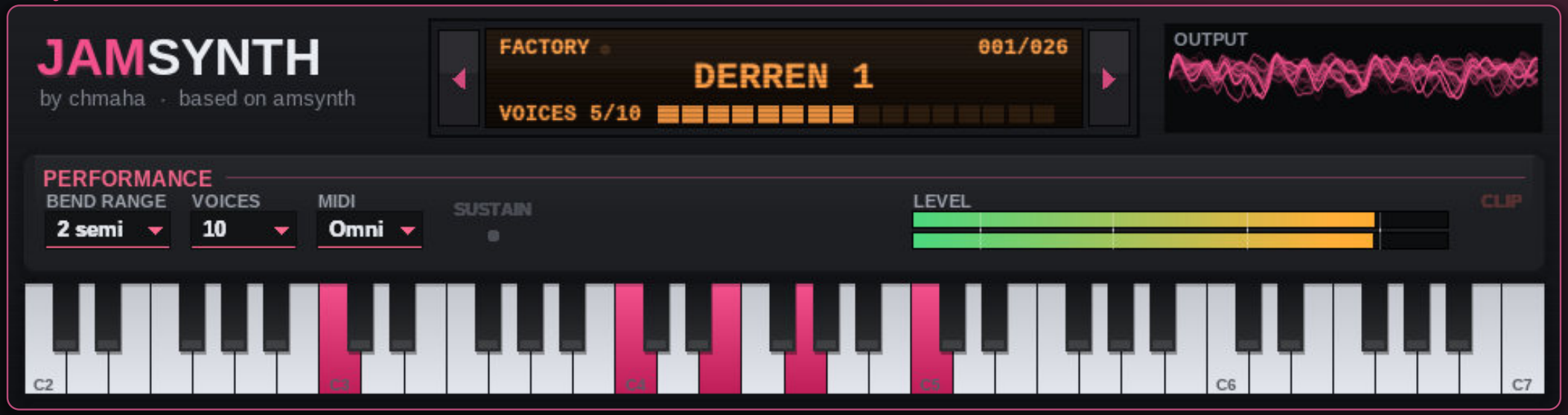
Based on amsynth



One screen, laid out as **OSC 1**, **OSC 2**, **FILTER**, **FILTER ENVELOPE**, **AMP ENVELOPE**, **LFO**, reverb, distortion and **KEYBOARD** (Poly, Mono or Legato, with glide). Pick a bank and preset from the browser. **STORE** saves your sound into the 128-slot user bank, where you can also **INIT** and rename slots. **JamSynth Presets** is the compact browser-only version (level, MIDI channel, polyphony and bend range).

JamSynth

JamSynth Presets





Jexed

A DX7-style FM synth with 1056 voices and a full voice editor.

Based on Dexed

Two pages: **PLAY** and **EDIT**. On PLAY, the cartridge shows 32 voice cards. Click a card to select it and edit it, or click the cartridge name to switch to another one (Dexed_01 and SynprezFM_01–32 are included). **STORE** saves the current voice into a user cartridge and **INIT** resets it. On EDIT you shape each operator (envelope, level, key scaling, frequency, detune), the **ALGORITHM** and feedback, the **LFO & GLOBAL** settings and the **CONTROLLERS** (mod wheel, foot, breath, aftertouch routing to pitch, amp and EG bias). The PLAY page also has output level, MIDI channel, polyphony, cutoff and resonance.

Jexed Presets is the compact version: a cartridge browser plus output level, MIDI channel and polyphony, with no editor.

Jexed

EDIT



Jexed Presets



INSTRUMENTS / 11 OF 24

Kaminari

Fifteen taiko drums and hits on key pairs, made for rolling.

Based on the sfizz engine, playing S. Christian Collins' SCC Taiko Drums



Fifteen drums and hits on key pairs from C2 to A#4. Each pair is a white key and the black key above it, so you can roll between them. How long you hold a key sets how long the drum rings (use the sustain pedal for full decays). Click a drum in the **DRUMS** list to hear it. The **CONTROLS** panel chooses how dynamics work: **Performance** (a controller such as the mod wheel, breath, foot or expression, plus velocity) or **Notation** (velocity only). **SETTINGS** holds **ENGINE** options (sample quality and polyphony) and tuning (concert pitch and temperament).

INSTRUMENTS / 12 OF 24

Lord of the Rinks

A tonewheel organ with drawbars, percussion and a rotary speaker.

Based on setBfree



Set the drawbars for **UPPER**, **LOWER** and **PEDALS** by dragging them. Panels cover **KEYBOARD & LEVELS** (MIDI channel per manual, split points, transpose), **PERCUSSION & VIBRATO** (volume, decay, harmonic, vibrato/chorus type), **OVERDRIVE · REVERB**, **ROTORS** (the Leslie, with horn and drum Stop/Slow/Fast, brake and acceleration), **MICROPHONES & CABINET** and **FILTERS**, and **DETAILS** (tonewheel and key-click modelling; changes to the tone generator are applied when you release the knob). **STORE** saves the panel as a program; type a name and press Enter. **MIDI LEARN** maps a hardware controller to a control: choose a control, then move the controller.

Major General MIDI

A 16-channel General MIDI wavetable synthesizer.

Based on Sonivox EAS



A 16-channel General MIDI synth. The **CHANNELS** panel sets the instrument, level, pan and reverb/chorus sends for each part (channel 10 is the drum kit). **FOLLOWS** makes one channel play another's instrument for layering. **REVERB** and **CHORUS** offer presets, and **CC91 SENDS / CC93 SENDS** or **WHOLE MIX** choose whether they follow the per-channel sends or process the whole mix. Other controls: **MASTER** volume, **TUNING** and **TEMPERAMENT** (drums are never retuned), and **GM RESET**. Over the A4 tuning field, the wheel changes pitch by 1 Hz.

Out of This Wurld

A sampled Wurlitzer EP200 reed piano with half-pedalling.

Based on the sfizz engine, playing Greg Sullivan's Wurlitzer EP200 samples



A sampled Wurlitzer EP200 electric piano, using Greg Sullivan's samples. Four velocity layers, with the sustain loops stored in the samples. Drag or scroll a knob (Ctrl for fine control) and double-click to reset it. **VOLUME** sets the output level (plus or minus 12 dB), **PAN** (CC10) the position in the stereo field, **RELEASE** (CC72) how long notes take to fade after the key comes up (0 to 2 seconds) and **VELOCITY** (CC99) how much key velocity changes the level (0% plays every note at full level). The **SUSTAIN** lamp glows as the dampers lift: click it to hold or release, or scroll or drag it for half-pedal. Pedalling from MIDI (CC64) works the same way, and pressing the pedal again catches notes that are still dying away. Click the keyboard to play (64 keys, A1 to C7); lower on a key is louder. Needs the Dependencies/OutOfThisWurld files; a bar shows the samples loading.



Partiality

Additive synthesis with up to eight voices per note and 141 ready-made sounds.

Based on the ADsynth engine of ZynAddSubFX

An additive synthesizer: every note is built from up to eight voices, each with its own stack of harmonics, and any voice can be shaped by a modulator. Pick a sound from the folders in the patch menu (the **Factory** folder holds the 19 hand-made sounds, the other folders hold 122 instruments from the ZynAddSubFX banks). The **Quick controls** (including **Punch**) are the fastest way to reshape a sound; for deeper editing, each voice has cards for its **Waveform**, **Harmonics**, **Modulator**, **Unison**, **Filter**, **Envelope** and **LFO** settings. Drag or scroll a knob (hold Ctrl for fine adjustment) and double-click to reset it. Set the **MIDI Channel** and **Master Volume** in the sliders. Edited sounds are saved with the project and in REAPER presets. The instrument banks need the Dependencies/Partiality files.



Plait Tectonics

24 synthesis models in one macro oscillator.

Based on Mutable Instruments Plaits

A macro oscillator in the style of Mutable Instruments Plaits. Click a tag in the middle to pick one of 24 models: the orange **NEW** bank (VA with filter, phase distortion, three DX7-style FM banks, wave terrain, string machine, chiptune), the teal **PITCHED** bank (virtual analog, wavefolder, FM, granular formants, additive, wavetable, chords, speech) and the pink **NOISE · DRUMS** bank (swarm, filtered noise, particles, string, modal resonator, kick, snare, hi-hat). The display shows what **HARMONICS**, **TIMBRE** and **MORPH** do in the selected model. Drag or scroll a knob (hold Ctrl for fine adjustment) and double-click to reset it.

FREQUENCY fine-tunes within the **RANGE** (Full sweeps C0-C5 with the knob; the C0-C7 settings put MIDI notes at pitch with ±7 semitones of fine tuning). The three small knobs set how much each note's decay envelope moves **TIMBRE**, the pitch (**FM**) and **MORPH**. **PING** plucks the low-pass gate on each note, **GATE + VEL** holds it open for as long as the key is down with velocity as the level, and **DRONE** plays continuously with MIDI notes only setting the pitch. **DECAY** and **COLOUR** shape the low-pass gate (drums, strings and the FM patches have their own envelopes). **OUTPUT** picks the main OUT, the alternative AUX output or both in stereo. Click the keyboard to audition. Pitch bend covers ±2 semitones. Needs the Dependencies/PlaitTectonics files.



Ring Leader

A resonator that turns notes or audio into strings, bells and plates.

Based on Mutable Instruments Rings

A resonator in the style of Mutable Instruments Rings. **MODEL** picks the resonator: **MODAL** (bars, bells, plates), **SYMPATHETIC** strings, **INHARMONIC** string, **FM VOICE**, **QUANTIZED** sympathetic strings tuned to a chord, and **STRING + VERB**. **VOICES** sets 1, 2 or 4 voices; each new note strums the next voice and the voices alternate between the left and right outputs. **FREQUENCY** transposes MIDI notes (in Audio Only mode it sets the pitch), **STRUCTURE** sets the harmonic structure (the chord in the quantized model, the ratio for the FM voice), **BRIGHTNESS** the tone, **DAMPING** the decay (fully right sustains) and **POSITION** the excitation point (the feedback for the FM voice). Drag or scroll a knob (hold Ctrl for fine adjustment) and double-click to reset it.

INPUT: **MIDI · INTERNAL** plucks or strikes the resonator on each note; **MIDI · AUDIO IN** feeds the track's audio into it, with notes setting the pitch and strumming; **AUDIO ONLY** makes it an effect, where transients in the audio strum it. Notes of a chord strum 10 ms apart, as on the module. Pitch bend covers ±2 semitones. Click the keyboard to audition. **ODD/EVEN** sends odd partials (or odd voices) left and even ones right, like the module's two outputs; **MIX** puts both on both channels.

DISASTROUS PEACE switches to the module's hidden string synth: each note plays a chord picked by **CHORD**, **REGISTRATION** chooses the octaves and waveforms, **ENVELOPE** goes from plucked to slow swells (fully right drones) and **FX AMOUNT** drives the **EFFECT**: formant, chorus, reverb, formant 2, ensemble or reverb 2.



Scoring Synth and Sound Blasted!

Two 16-channel SoundFont synths with reverb, chorus and historical temperaments.

Based on FluidSynth, with GeneralUser GS and HL4MGM

SoundFont synths (GeneralUser GS and HL4MGM) with 16 multitimbral channels. The **CHANNELS** panel sets the bank and program, level, pan and effect sends for each channel, plus **FOLLOWS** layering. Click a channel number to choose which channel the on-screen keyboard plays. Global controls: **REVERB** (engine **FDN**, **Freeverb** or **Dattorro**, with room size, damp, width and level), **CHORUS**, **MASTER** volume, polyphony, **INTERPOLATION** and **NOTE CUT**. A **MIDI MONITOR** shows incoming messages (the **NOTES** button shows or hides notes, **CLEAR** empties it), and **GM RESET** sends a reset. Tuning and temperament are as in Major General MIDI; hold Ctrl while wheeling over A4 for 0.1 Hz steps.

Scoring Synth and Sound Blasted!

Sound Blasted!

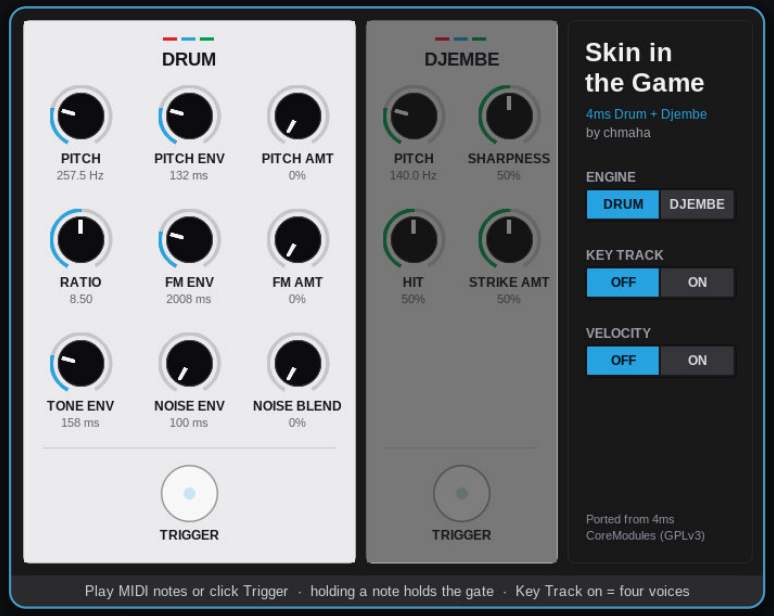


INSTRUMENTS / 19 OF 24

Skin in the Game

An FM drum and a modal djembe, played from MIDI.

Based on 4ms Drum and Djembe (CoreModules)
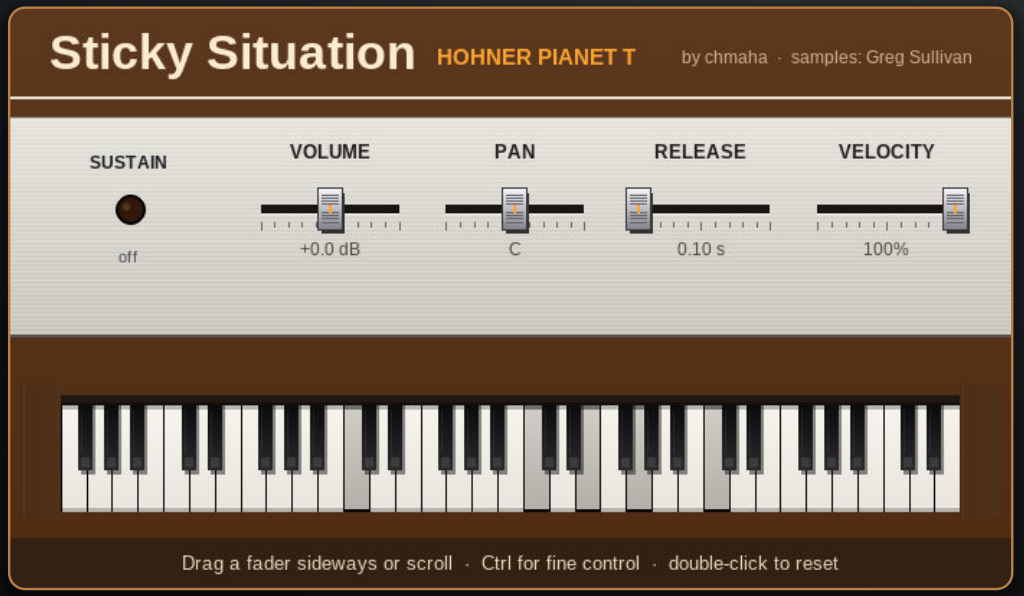


Two 4ms drum modules in one. Click the **DRUM** or **DJEMBE** panel, or the **ENGINE** switch, to choose which one plays; the other panel is dimmed. The **DRUM** panel is a two-operator FM drum: **PITCH** (10 Hz to 1 kHz), **PITCH ENV** and **PITCH AMT** (the pitch sweep at the start of each hit), **RATIO**, **FM ENV** and **FM AMT** (the modulator), **TONE ENV** (how long the body rings), and **NOISE ENV** and **NOISE BLEND** (a noise layer for snares and hats). The **DJEMBE** panel is a 20-mode resonant hand drum: **PITCH** (20 to 500 Hz), **SHARPNESS**, **HIT** and **STRIKE AMT**. Any MIDI note strikes the drum, and holding the note holds the gate (the Drum's envelopes sustain while held); **TRIGGER** does the same from the mouse. With **KEY TRACK** off every note re-strikes one voice at the Pitch knob; with it on the note sets the pitch (middle C plays the Pitch knob; the Djembe only goes up from there) and up to four hits ring at once. **VELOCITY** scales each hit's level by velocity; off, every hit plays at the module's own level. Drag or scroll a knob (Ctrl for fine control) and double-click to reset it.

INSTRUMENTS / 20 OF 24

Sticky Situation

A sampled Hohner Pianet T, with key-release samples and half-pedalling.

Based on the sfizz engine, playing Greg Sullivan's Pianet T samples



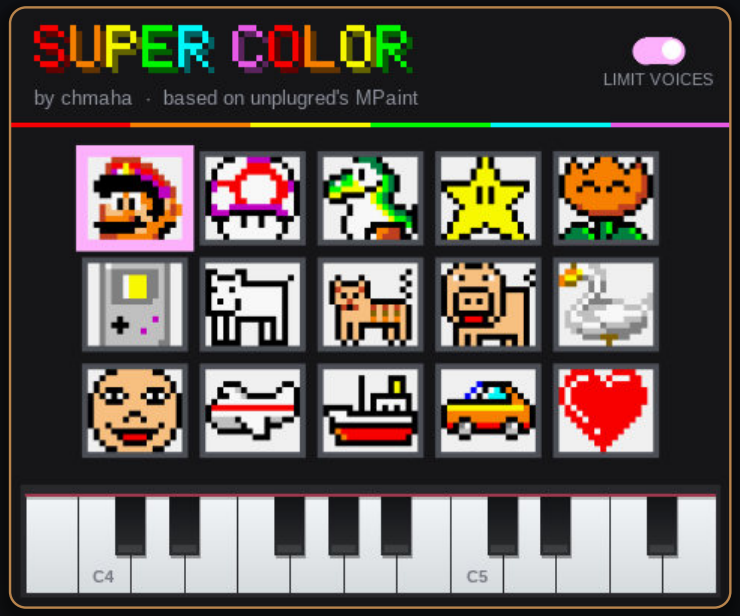
A sampled Hohner Pianet T reed piano, using Greg Sullivan's samples. Two velocity layers, plus key-release samples that get quieter the longer a key is held. The Pianet T had no controls of its own, so they sit on faders: drag one sideways or scroll it (Ctrl for fine control) and double-click to reset it. **VOLUME** sets the output level (plus or minus 12 dB), **PAN** (CC10) the position in the stereo field, **RELEASE** (CC72) how long notes take to fade after the key comes up (0 to 2 seconds) and **VELOCITY** (CC99) how much key velocity changes the level (0% plays every note at full level). The **SUSTAIN** lamp glows as the dampers lift: click it to hold or release, or scroll or drag it for half-pedal. Pedalling from MIDI (CC64) works the same way, and pressing the pedal again catches notes that are still dying away. Click the keyboard to play (60 keys, F1 to E6); lower on a key is louder. Needs the Dependencies/StickySituation files; a bar shows the samples loading.

INSTRUMENTS / 21 OF 24

Super Color

A little sampler with fifteen sounds, played from B3 to G5.

Based on unplugred's MPaint



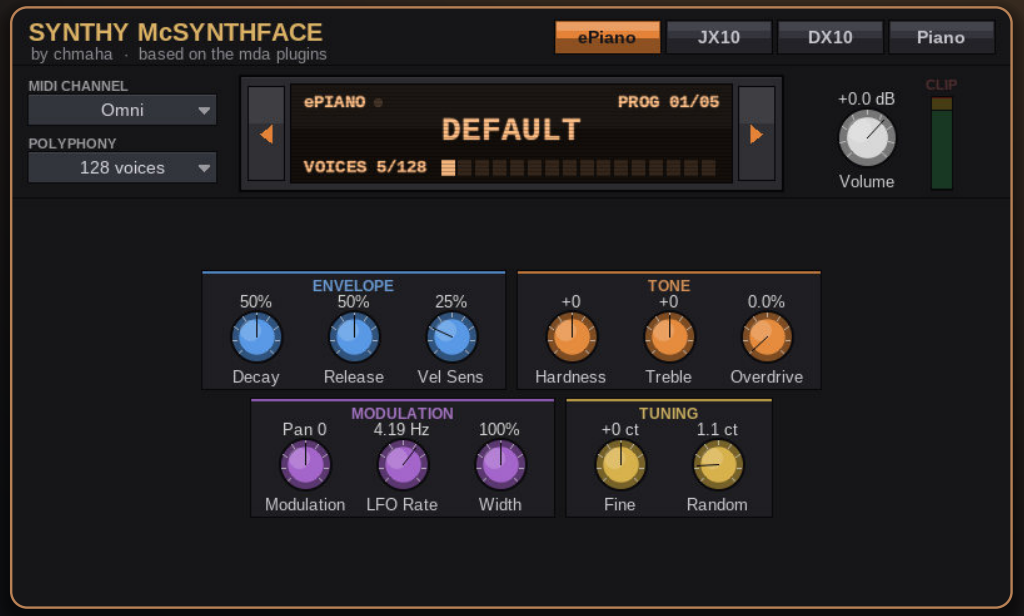
Click an icon in the grid to choose a sound (fifteen in all). Play it from MIDI between B3 and G5, or click and drag across the on-screen keyboard. **LIMIT VOICES** switches the original's monophonic/three-voice note handling on or off.

INSTRUMENTS / 22 OF 24

Synthy McSynthface

Four instruments in one: Piano, ePiano, JX10 and DX10.

Based on the mda plugins



Four instruments in one. Click a tab to choose **Piano**, **ePiano**, **JX10** or **DX10**. A display shows the current program name and MIDI activity; step through programs there. Each tab has its own panels of knobs (Ctrl-drag for fine adjustment). Set the **MIDI CHANNEL** and **POLYPHONY** in the boxes at the top, and watch the master volume meter's **CLIP** light.

Tine Capsule

A sampled 1977 stage tine piano with tremolo, autopan and half-pedalling.

Based on the sfizz engine, playing Jeff Learman's jRhodes3d samples

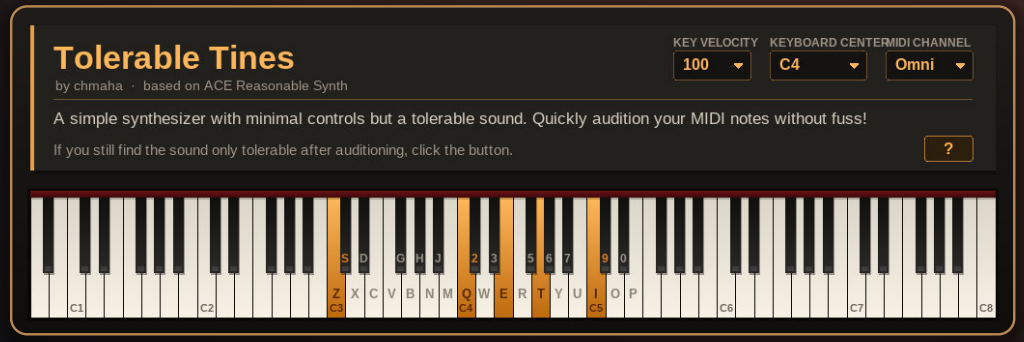


A sampled 1977 Mark I Stage 73 tine piano, using Jeff Learman's jRhodes3d samples. **VOLUME** sets the output level (plus or minus 12 dB). The **TREMOLO** knobs are **Depth** (mod wheel, CC1, up to 6 dB), **Rate** (CC92, 0.1 to 10 Hz), **Delay** (CC93, how long after each note it starts) and **Fade** (CC94, how long it takes to fade in); Delay and Fade act on the next notes. **AUTOPAN** has **Depth** (CC95) and **Rate** (CC96). **VELOCITY CROSSFADE** blends the five velocity layers into each other, or switches hard between them when off (no phasing on low notes). **KEYS** chooses 73 keys (E1 to E7, like the sampled instrument; other notes are silent) or 88 keys (A0 to C8). The **SUSTAIN** lamp glows as the dampers lift: click it to hold or release, or scroll or drag it for half-pedal. Pedalling from MIDI (CC64) works the same way, and pressing the pedal again catches notes that are still dying away. Click the keyboard to play; lower on a key is louder. Drag or scroll a knob (Ctrl for fine control) and double-click to reset it. Needs the Dependencies/TineCapsule files; the first load takes about a second.

Tolerable Tines

An electric piano with six hidden ear-training games.

Based on Ardour's ACE Reasonable Synth



An electric piano with an 88-key on-screen keyboard that you can also play from the computer keyboard. A **MIDI CHANNEL** box filters incoming MIDI and **KEY VELOCITY** sets the velocity of the on-screen keys. The default sound is *Tolerable*. If you find it merely tolerable, click the button to switch to the *Mystery* sound, which changes slightly every day. The plugin also hides six ear-training keyboard games (Name That Note, Higher or Lower?, Interval Spotter, Chord Detective, Simon Plays and a scale game) with Easy to Extreme levels, and a December lights display that plays Jingle Bells when clicked.



EFFECTS

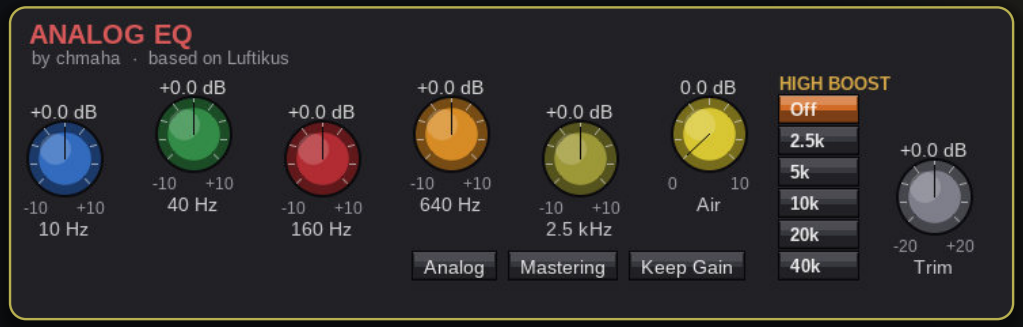
EQ, dynamics, tape, space and some stranger things for the tracks you already have.

28 ENTRIES

Analog EQ

Five broad bands and an air shelf.

Based on Luftikus (lkjb)



Five band knobs (10 Hz, 40 Hz, 160 Hz, 640 Hz, 2.5 kHz) cut or boost up to 10 dB. **Air** adds a high shelf, with a corner set from 2.5 kHz to 40 kHz. **Analog**, **Mastering** and **Keep Gain** are switches, and **Trim** sets the output level.

Analog Jam

Tape stop and tape start, on cue.

Based on cStop



A tape stop and tape start. **Automate the Mode parameter** (Bypass, Stop or Start) to stop or start the tape; the display shows the state (running, stopping, stopped or starting) and the current speed. **SLOWDOWN** and **SPEEDUP** each have a length, curve, start and end. Turn **SYNC** on to set the lengths in bars (1/32 up to 8) from the host tempo. **FADE** and **XFADE** smooth the transitions, **AUTO BYP** returns to bypass automatically, and **FILTER** (low, high or band pass, **CUTOFF**, **RESO**) colours the tape.

Autotuner

Pulls pitch to the notes you choose, from a scale, the keyboard or live MIDI.

Based on x42 fat1 (zita-at1)



Notes From picks the target notes: **Auto** uses the keyboard (or held MIDI notes when there are any), **MIDI** uses only held MIDI notes, **Manual** uses only the keyboard. Click keys to enable or disable notes, or choose a **Scale**. **MIDI Channel** filters incoming notes and pitch bend. Knobs: **Tuning (A4)**, **Bias** (preference for staying on the current note), **Filter** (how smoothly the correction follows), **Correction** (amount), **Offset** (±2 semitones) and **Bend Range** for MIDI pitch bend. **Fast Correction** cuts the latency to an eighth. The bottom row detunes each note by up to ±100 cents for microtonal scales. The meter shows the current note and its pitch error.



them and the SIZE modulation knob becomes a low-pass (left) or high-pass (right) filter. **FREEZE** stops recording; in delay mode it turns the buffer into a beat slicer where TIME picks the slice.

FEEDBACK, **DRY/WET** and **REVERB** sit on the sides. **QUALITY** chooses the buffer: clean digital (4 seconds), vintage digital (8 s, 12-bit at 32 kHz), clean tape (10 s) or dirty tape (16 s, 8-bit mu-law at 24 kHz with wow and flutter). Mono input doubles the buffer length. Changing quality, input or source clears the buffer with a short silence, and the change waits while the buffer is frozen. The input gain is manual (0 to +32 dB, soft-clipping on the tape settings) or automatic, as on the module.

With **SOURCE** set to **WAVETABLE**, it becomes the module's built-in synth: FEEDBACK picks one of 12 wavetable banks, TIME (WAVE) moves through the bank, PITCH tunes it and DRY/WET balances the plain oscillator against a granular cloud of the same waves. This needs Dependencies/BeadTimeStory/beads_waves.txt. The display shows the buffer with grains as dots and delay taps as lines, or the current wave and the output in synth mode.

Bead Time Story

Turns a track's audio into a cloud of grains.

Based on Mutable Instruments Beads

A granular texture synthesizer in the style of Mutable Instruments Beads. It records the track's audio into a buffer and replays it as a cloud of grains. **DENSITY** sets how grains are seeded: at noon nothing happens, to the left grains come at a regular rate, to the right at random. **TIME** chooses where in the buffer grains start (left: the most recent audio), **SIZE** their length (left of 9 o'clock plays them backwards), **SHAPE** their envelope and **PITCH** their transposition, with detents at musical intervals. The small knob under each one adds random modulation: to the right every grain gets a random value, to the left only some of them do. Drag or scroll a knob (hold Ctrl for fine adjustment) and double-click to reset it.

SEED seeds grains by hand. When **LOCK** is on (the button glows), grains flow on their own; pressing SEED unlocks it, so grains only flow while you hold it. Hold SEED for 1.5 seconds, or click LOCK, to lock it again. With **MIDI NOTES** on **SEED**, notes act as the module's SEED input. Locked, they become a clock that DENSITY divides (left of noon) or thins out at random (right). Unlocked, grains flow while a note is held. **+PITCH** also transposes from the notes (C4 leaves the pitch alone).

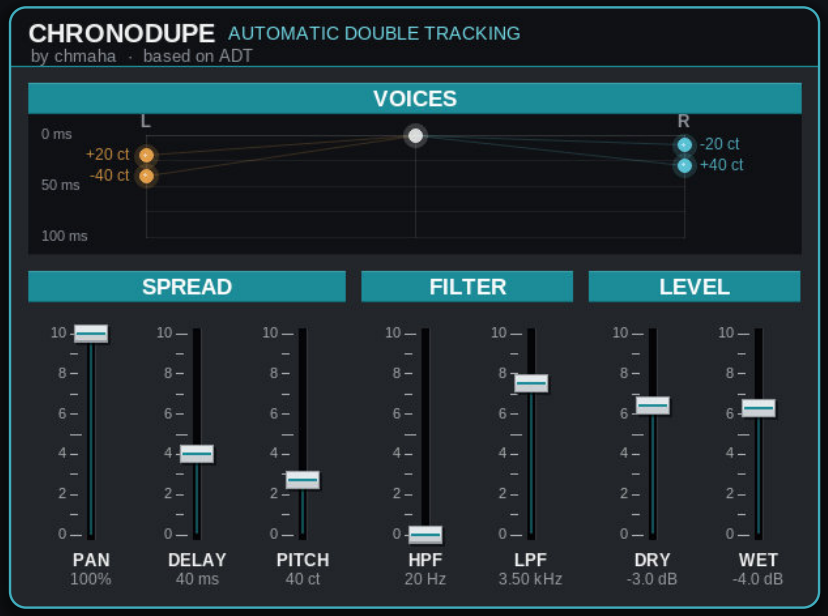
Turn **SIZE** fully clockwise (∞) and the module crossfades into a multi-tap delay. With no MIDI clock and LOCK on, DENSITY sets the delay time (longest at noon) and past noon fades in extra taps at odd ratios. Tap SEED twice, or send MIDI notes, to set a tempo, and DENSITY then picks multiples and divisions of it. TIME multiplies the delay time, SHAPE gates or swells the taps, PITCH pitch-shifts

EFFECTS / 5 OF 28

ChronoDupe

Automatic double tracking.

Based on ADT (Christopher Arndt)



Automatic double tracking. **VOICES** has the **PAN**, **DELAY** and **PITCH** spread; **FILTER** has the **HPF** and **LPF**; **LEVEL** sets **DRY** and **WET**.

EFFECTS / 6 OF 28

Compressor

A compressor with a live transfer curve or level history.

Based on x42 darc



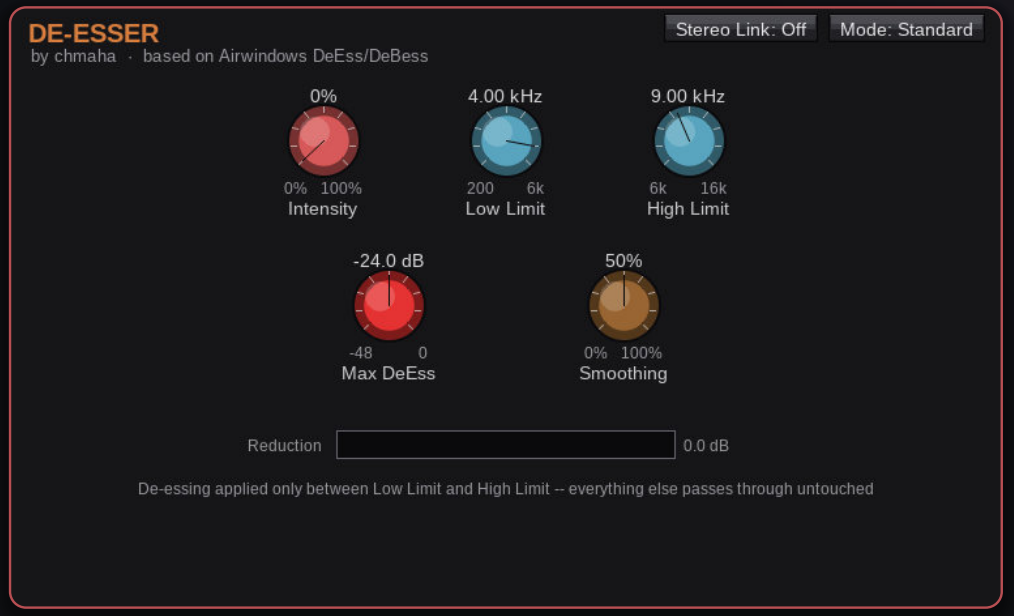
Knobs: **Input Gain**, **Threshold**, **Ratio** (**Lim** at maximum), **Attack** and **Release**. **Hold** keeps the gain reduction steady. The graph button switches between **Curve** (input to output) and **History** (level and gain reduction scrolling from right to left).

EFFECTS / 7 OF 28

De-Esser

Tames sibilance, and lets you hear what it removes.

Based on Airwindows DeEss and DeBess



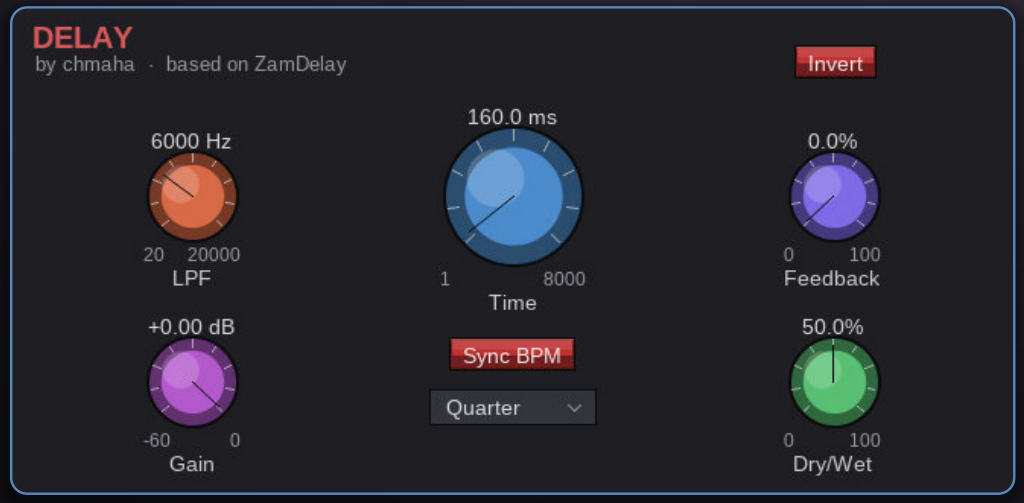
Choose **Mode: Standard** or **Advanced**. Knobs: **Intensity**, **Low Limit**, **High Limit** and **Max DeEss**. Advanced mode adds **Smoothing**, **Sharpness**, **Depth** and **Filter**. **Stereo Link** ties both channels together and **Monitor: Delta** lets you hear what is being removed. A **Reduction** meter shows the effect.

EFFECTS / 8 OF 28

Delay

A straightforward delay with tempo sync.

Based on ZamDelay



Knobs: **Time**, **Feedback**, **LPF**, **Dry/Wet** and **Gain**. Click **Sync BPM** to set the time by note value instead (Whole, Half, Quarter, Eighth or Sixteenth). **Invert** flips polarity.

EFFECTS / 9 OF 28

Dizzy

A rotary speaker with separate horn and drum.

Based on b_whirl from setBfree



Horn and drum columns, each with **STOP / SLOW / FAST** speed, **BRAKE**, **ACCEL**, **DECEL** and level. A link setting makes one rotor follow the other or run independently. The **FILTERS** section shapes each rotor, and the microphone section sets **ANGLE**, **DISTANCE**, and the horn and drum **RADIUS**. The header switches between **ENABLED** and **BYPASSED**.

EFFECTS / 10 OF 28

Dynamic Tilt

A tilt EQ whose slope follows the signal.

Original design



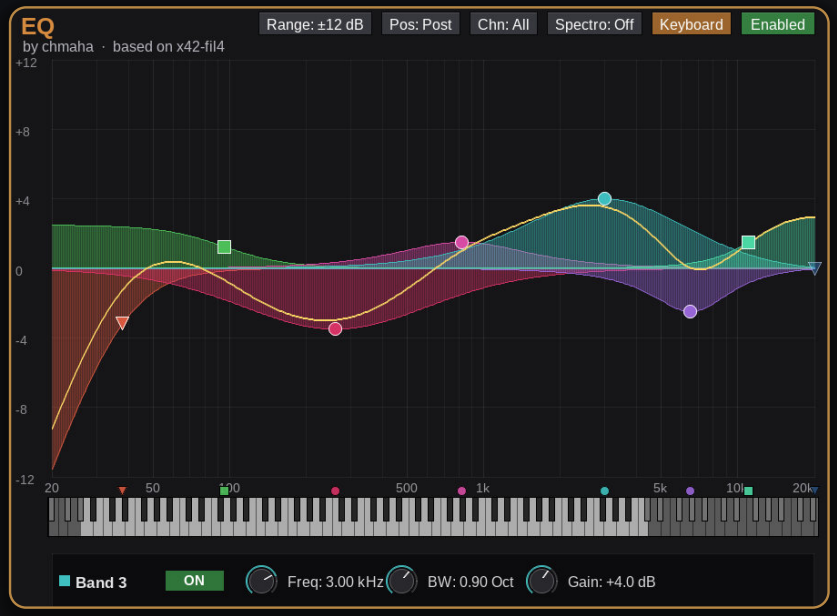
A tilt EQ pivoting at **Pivot**. The detector is either **Level** or **Brightness**, each with its own **Threshold**, **Amount** or **Ratio**, **Max Tilt** or **Range**, **Attack** and **Release**. In Brightness mode, **Learn** averages the input's brightness to set the threshold for you, and **Two-Sided** lets the tilt go in either direction. **Auto Gain** keeps the loudness steady and **Output** trims the result.

EFFECTS / 11 OF 28

EQ

Eight nodes, set directly on the graph.

Based on x42 fil4



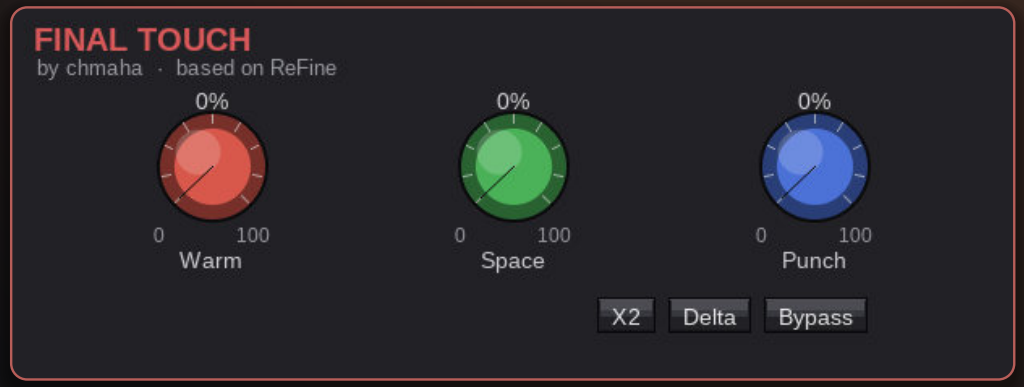
Click a node on the graph to select it, then adjust its frequency, gain and bandwidth (or resonance for the high and low pass) with the controls underneath. Nodes: highpass, low shelf, bands 1–4, high shelf and lowpass. The mouse wheel over a node changes its bandwidth. Buttons set the enable state, **Chn** (all, left or right), **Pos** (spectrum before or after the EQ), **Spectro** mode, **Spd** (speed), **Res** (resolution) and the graph **Range**.

EFFECTS / 12 OF 28

Final Touch

Warm, Space and Punch.

Based on ReFine (lkjb)



Three knobs: **Warm**, **Space** and **Punch**. Header buttons: **X2** (doubles the effect), **Delta** (hear only what is being added) and **Bypass**.

EFFECTS / 13 OF 28

Gate

A gate that can also duck, with sidechain.

Based on ZamGateX2



Knobs: **Threshold**, **Attack**, **Release**, **Range** (all the way down is **Mute**) and **Makeup**. Header buttons: **Mode: Gate / Duck** and **Sidechain: On / Off**.

EFFECTS / 14 OF 28

Inflator

Four knobs of extra loudness and density.

Based on RCInflator2



Knobs: **Input**, **Effect**, **Curve** and **Output**. Buttons: **Effect In**, **Band Split** and **Clip 0 dB**. Peak and RMS readouts show the level.

EFFECTS / 15 OF 28

Intelligent Spectral Compressor

A compressor working on every frequency band.

Based on Spectral Compressor



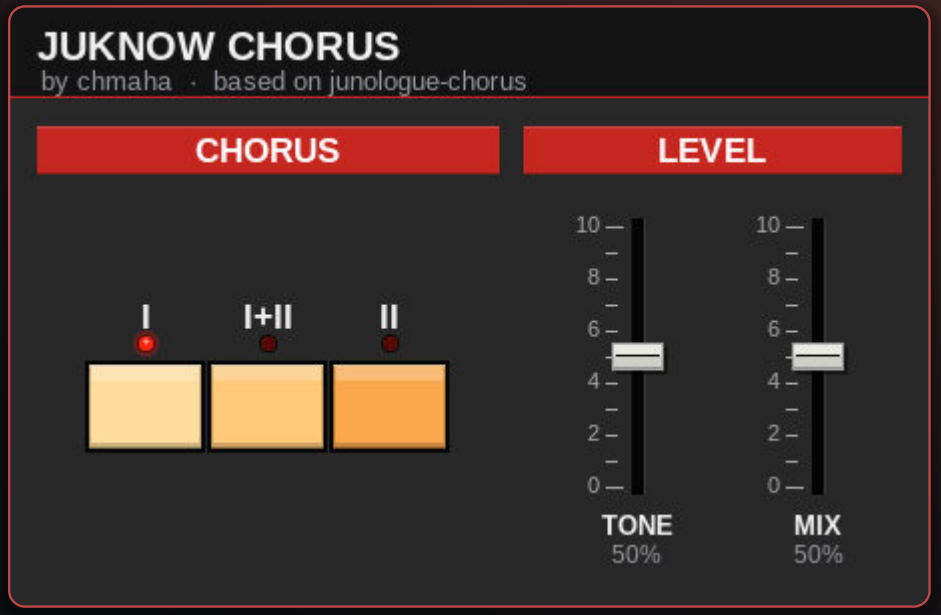
A compressor working on every frequency band. The **Easy** view offers **Tame** (downward ratio), **Recover** (upward ratio), **Tilt** (slope), **Mix** and **Output**. **Advanced** exposes everything: **Threshold**, **Center**, **Slope**, **Curve**, **Mode** (**Pink Noise**, **SC Match** or **SC Comp**), **SC Link**, **Attack**, **Release**, **Window**, **Overlap**, and **Down** / **Up** offset, ratio, rolloff and knee. **Auto Threshold** and **Auto Balance** help set levels.

EFFECTS / 16 OF 28

JuKnow Chorus

I, I+II or II. Pick a mode, set the tone.

Based on junologue-chorus (Peter Allwin)

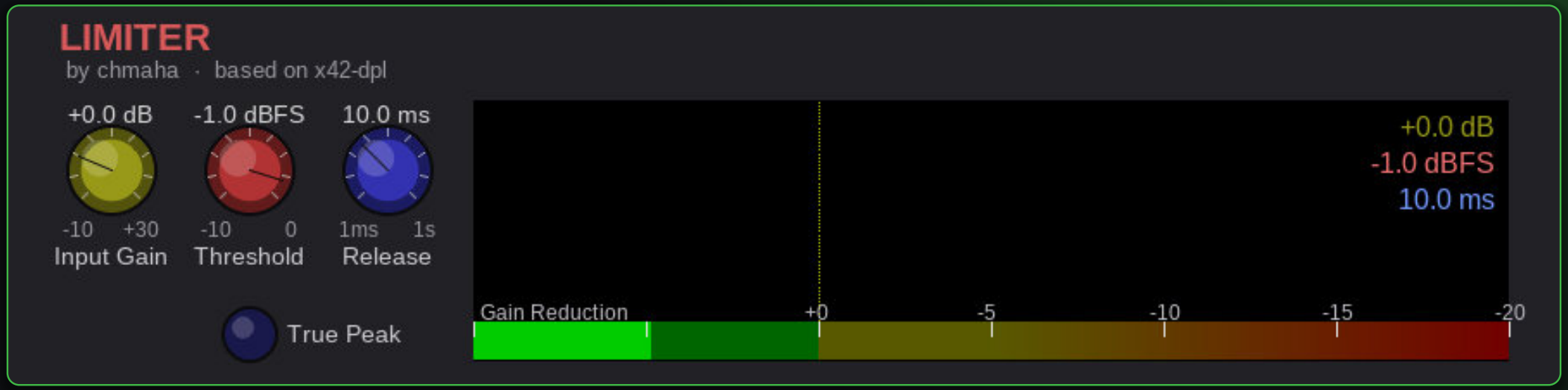


Click **I**, **I+II** or **II** for the chorus mode, then set **TONE** and **MIX**.

Limiter

A limiter with optional inter-sample peak limiting.

Based on x42 dpl

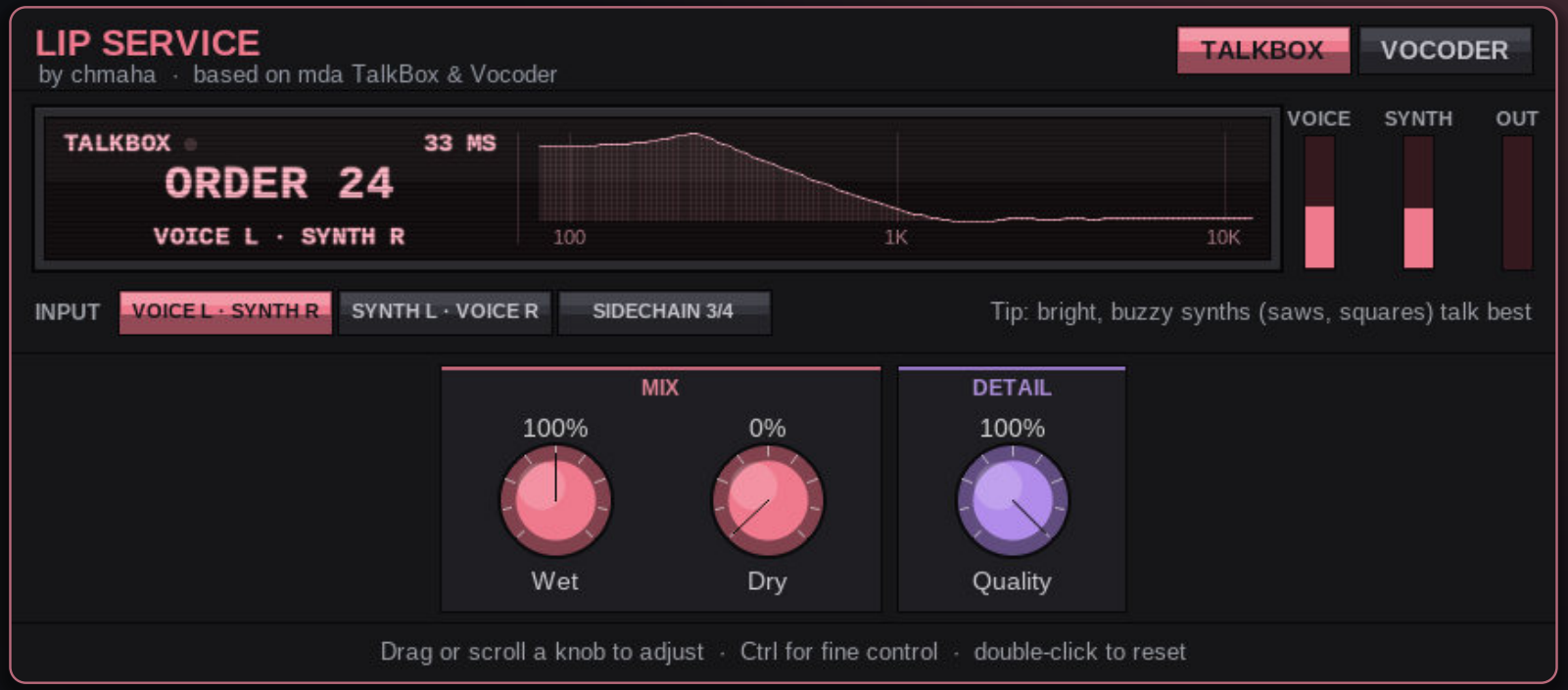


Knobs: **Input Gain**, **Threshold** and **Release**, with a **True Peak** button for inter-sample peak limiting. The meter shows **Gain Reduction**.

Lip Service

Make a synth talk, with a talk box or a vocoder.

Based on mda TalkBox and Vocoder



Makes a synth talk. The voice and the synth come in on separate inputs: by default the voice is the track's own audio (channels 1+2) and the synth comes in on the sidechain, so send a synth track to this track's channels 3/4. **INPUT** can instead take a stereo signal with the voice on one side and the synth on the other (**VOICE L · SYNTH R** or **SYNTH L · VOICE R**). Bright, buzzy synths (saws, squares, pads) talk best. The **TALKBOX** and **VOCODER** buttons at the top pick the mode.

TalkBox models your voice like a talk box tube in the mouth. **MIX**: **Wet** sets the level of the talking synth and **Dry** mixes the unprocessed voice back in (both up to 200%). **DETAIL**: **Quality** sets how much detail the voice model keeps; lower sounds rougher and more muffled. The display shows the model's order (24 at full Quality and 48 kHz, fewer at lower settings) and the voice's formant curve.

Vocoder is a classic channel vocoder. **FILTER BANK**: **Mid Freq** shifts the whole bank up or down to change the voice character, **Filter Q** sets how sharp and ringing the bands are, and **8/16** picks the number of bands (8 for a gritty vintage sound, 16 for clearer speech). **ENVELOPE**: **Envelope** sets how fast the bands follow the voice; fully left freezes the current sound. **HIGHS**: **Hi Band** adds the synth's highs, shaped by the voice, and **Hi Thru** lets the voice's own highs (s, t, ch) through for clearer words. **OUTPUT** sets the level. **PRESET** recalls **CLASSIC**, **16 BAND**, **OLD**, **CHORAL** and **PAD**; the button lights while the settings match. The display shows the band levels.

The **VOICE**, **SYNTH** and **OUT** meters show the levels, and the display warns **NO SYNTH SIGNAL** when the voice is playing but nothing reaches the synth input.

Lip Service


LIP SERVICE
by chmaha · based on mda TalkBox & Vocoder
TALKBOX
VOCODER
VOCODER
8 BANDS
800 HZ
VOICE L · SYNTH R
190 HZ
3000 HZ
HI
VOICE
SYNTH
OUT
INPUT
VOICE L · SYNTH R
SYNTH L · VOICE R
SIDECHAIN 3/4
PRESET
CLASSIC
16 BAND
OLD
CHORAL
PAD
FILTER BANK
8
16
800 Hz
55%
Mid Freq
Filter Q
ENVELOPE
30 ms
Fast
Slow
Envelope
HIGHS
40%
40%
Hi Band
Hi Thru
OUTPUT
+0.0 dB
Output
Drag or scroll a knob to adjust · Ctrl for fine control · double-click to reset

EFFECTS / 19 OF 28

Lumina Channel Strip

Filters, gate, EQ, compressor, saturation and limiter in one plugin.

Based on the ports in this collection



A full channel strip. From input to output: **INPUT** (gain, **PHASE INV**), **FILTER** (high and low pass with **FREQ** and **RES**), **GATE**, an analog **EQ** with presets (Flat, Acoustic, Bass, Classical, Electronic, Pop, Rock, Vocal), **COMPRESSOR** (threshold, ratio, attack, release, **HOLD**, **SC HPF**, **PARALLEL**), **SATURATION** (the five consoles, **DRIVE**, **MIX**), **LIMITER** and **TRUE PEAK** metering. Each section can be switched on and off, and **PRE EQ / POST EQ** sets where the EQ sits relative to the compressor.

EFFECTS / 20 OF 28

Maximizer

Release, Threshold and Ceiling.

Based on ZaMaximX2



Three knobs: **Release**, **Threshold** and **Ceiling**. The display shows gain reduction and output level.

EFFECTS / 21 OF 28

Multiband Compressor

Three bands, each with its own compressor.

Based on ZaMultiCompX2



Three bands: **LOW**, **MID** and **HIGH**. Click a band to edit it (the header shows which band is being edited), then set its **Threshold**, **Ratio**, **Knee**, **Attack**, **Release** and **Makeup**. **Solo** auditions a band. **Xover 1** and **Xover 2** set the split frequencies, **Master Trim** sets the output level, and **Detect: Avg / Max** picks the detector.

EFFECTS / 22 OF 28

Potato Quality

Starts clean, so you can dial the damage in.

Based on mda Degrade



A lo-fi degrader that starts clean, so nothing changes until you turn a knob. **CRUSH**: **Quant** sets the bit depth (24 down to 4 bits) and **Rate** lowers the sample rate from the centre outwards; turn it toward **Hold** for gritty, aliased single samples or toward **Avg** for smoother averaged ones. **SATURATE**: **Non-Lin** adds harmonic thickening (**Even** to the left, **Odd** to the right) and **Headroom** sets the clipping level. **FILTER**: **Post Filter** is a steep low-pass that muffles the result; turning it switches the filter on, and the **ON/OFF** button bypasses it. **MONO** sums the input like the original plugin; **STEREO** degrades each channel on its own. The display shows the bit depth, sample rate and a scope of the output, and **Output** sets the level next to the meter (click **CLIP** to reset it).



Reel Talk

A reel-to-reel tape machine, right down to the head gap.

Based on CHOW Tape Model (Jatin Chowdhury)

A reel-to-reel tape machine. The panels have tabs along the top; click a tab to switch pages, and use the power button at the bottom of a page to switch that section on or off. **Gain** has **Input Gain**, **Dry/Wet** and **Output Gain**; **Filters** has **Low Cut**, **High Cut** and **Makeup** (adds the filtered-out signal back after the tape); **Stereo** has **Mid/Side**, **Balance** and its **Makeup**. **Tape** sets **Bias**, **Saturation** and **Drive**, the heart of the sound; **Comp** adds tape-style compression (**Amount**, **Attack**, **Release**) and **Tone** has **Bass**, **Treble** and **Transition Freq**.

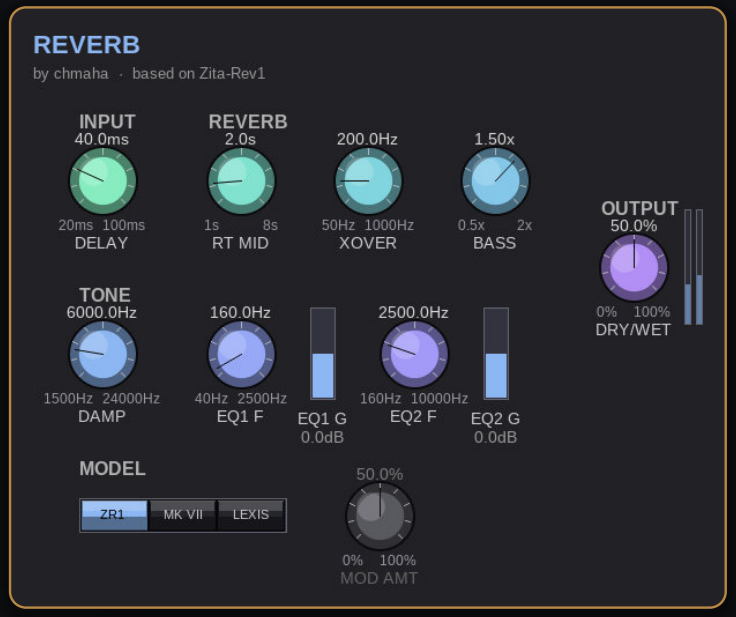
Loss models the playback head: **Gap**, **Thickness**, **Spacing**, **Azimuth** and **Speed**, with quick buttons for 3.75, 7.5, 15 and 30 ips. **Degrade** (**Depth**, **Amount**, **Variance**, **Envelope** and a **0.1x** range switch) adds noise and dulling, and **Chew** (**Depth**, **Frequency**, **Variance**) adds crinkled, damaged tape. **Flutter** and **Wow** set the pitch wobble (**Depth**, **Rate**, and for wow **Variance** and **Drift**); the **Flutter Sync** and **Wow Sync** buttons lock the rate to the tape speed or the project tempo.

Along the bottom, click **Oversampling** to pick 1x to 16x and minimum or linear phase, and **Tape Mode** to pick the solver: RK2 is the cheapest, NR8 the most accurate, STN uses a small neural network (it needs Dependencies/ReelTalk/stn_weights.txt) and V1 is the original v1 sound. Click the preset name for the factory presets, or use the arrows to step through them. The display at the top shows the input and output levels and a scope of the output, and the line at the bottom explains whatever is under the mouse.

Reverb

A reverb with three models: ZR1, MK VII and LEXIS.

Based on Zita-Rev1



Knobs: **DELAY** (pre-delay), **XOVER**, **BASS**, **RT MID** (decay time), **DAMP**, **TONE** (two-band EQ), **MOD AMT** and **DRY/WET**. **MODEL** chooses ZR1, MK VII or LEXIS.

Saturation

Five consoles, two knobs.

Based on Airwindows ChannelX



Choose a **Console** (Burnley, Maryland, Oxford, Tokyo or Washington), then set **Drive** and **Output**.

Silver Lining

Granular, stretch, looping delay and spectral textures.

Based on Mutable Instruments Clouds



A texture processor in the style of Mutable Instruments Clouds. It records the track's audio into a short buffer and plays it back. The teal icons pick the mode: **GRANULAR** (clouds of grains), **STRETCH** (pitch shifter and time stretcher), **LOOPING** delay and **SPECTRAL**; the black button next to them cycles through them. The pink icons pick the buffer quality: 16-bit stereo, mono, lo-fi stereo and lo-fi mono (longer buffers, grittier sound; changing quality clears the buffer). The green LEDs show the level. Drag or scroll a knob (hold Ctrl for fine adjustment) and double-click to reset it.

FREEZE stops recording so you can play with what is in the buffer. **POSITION**, **SIZE**, **DENSITY** and **TEXTURE** change meaning with the mode (the hint bar explains each one); **PITCH** transposes, with detents at musical intervals, and **IN GAIN** sets the input level. The **BLEND** knob controls whichever of **BLEND** (dry/wet), **SPREAD**, **FEEDBACK** and **REVERB** is selected with the pencil button or the chips under the display.

The display shows the buffer, newest audio at the right, with the grains or playheads on top (the spectrum in spectral mode). **TRIG**, or any MIDI note, fires the trigger: one grain, a tap tempo for the looping delay, a restart for the stretcher or a glitch in spectral mode. With **TRIG + PITCH**, MIDI notes also transpose (C4 leaves the pitch alone).

Warp Factor

Two signals in, one morphing signal out.

Based on Mutable Instruments Warps



A meta-modulator in the style of Mutable Instruments Warps: two signals go in and one comes out, combined through a continuously morphing set of algorithms. **ALGORITHM** sweeps from crossfade through wavefolding, analog and digital ring modulation, XOR and comparator to a 20-band vocoder, where the last part of the knob grows the release until it freezes. **TIMBRE** changes the character of the current algorithm (the formant shift in the vocoder), and **LEVEL 1** and **LEVEL 2** set the level and drive of each input. Drag or scroll a knob (hold Ctrl for fine adjustment) and double-click to reset it.

INPUTS: by default input 1 (the carrier) comes from channels 1+2 and input 2 (the modulator) from channels 3+4, so send a second track into channels 3/4; choose **L / R** to use the left and right channels of a normal track instead. **INT. OSC** replaces input 1 with the internal oscillator (click to cycle Off, Sine, Triangle and Saw; these become saw, pulse and noise in the vocoder), and LEVEL 1 then sets its pitch, or play it from MIDI. **OUTPUTS**: **MAIN** on both channels, or **MAIN + AUX** with the aux output on the right. **EASTER EGG** turns it into the module's hidden frequency shifter, with the knobs becoming feedback, dry/wet, shift (or phase rotation) and the blend of up and down shift. Levels are the module's own, so very hot signals saturate. Needs the Dependencies/WarpFactor files.

Wax Lyrical

Makes any audio sound like it is playing from an old record.

Based on Calf Vinyl (Markus Schmidt)



Makes any audio sound like it is playing from an old record. Knobs: **Input** and **Output** gain, **Drone** (a slow, off-centre pitch wobble that turns once per revolution of the record), **Speed** (the record speed, with quick **33**, **45** and **78** buttons beneath it), **Aging** (narrows the frequency range the way a worn record loses its lows and highs) and **Frequency** (the centre of that honky midrange). Drag or scroll a knob (hold Ctrl for fine adjustment) and double-click to reset it.

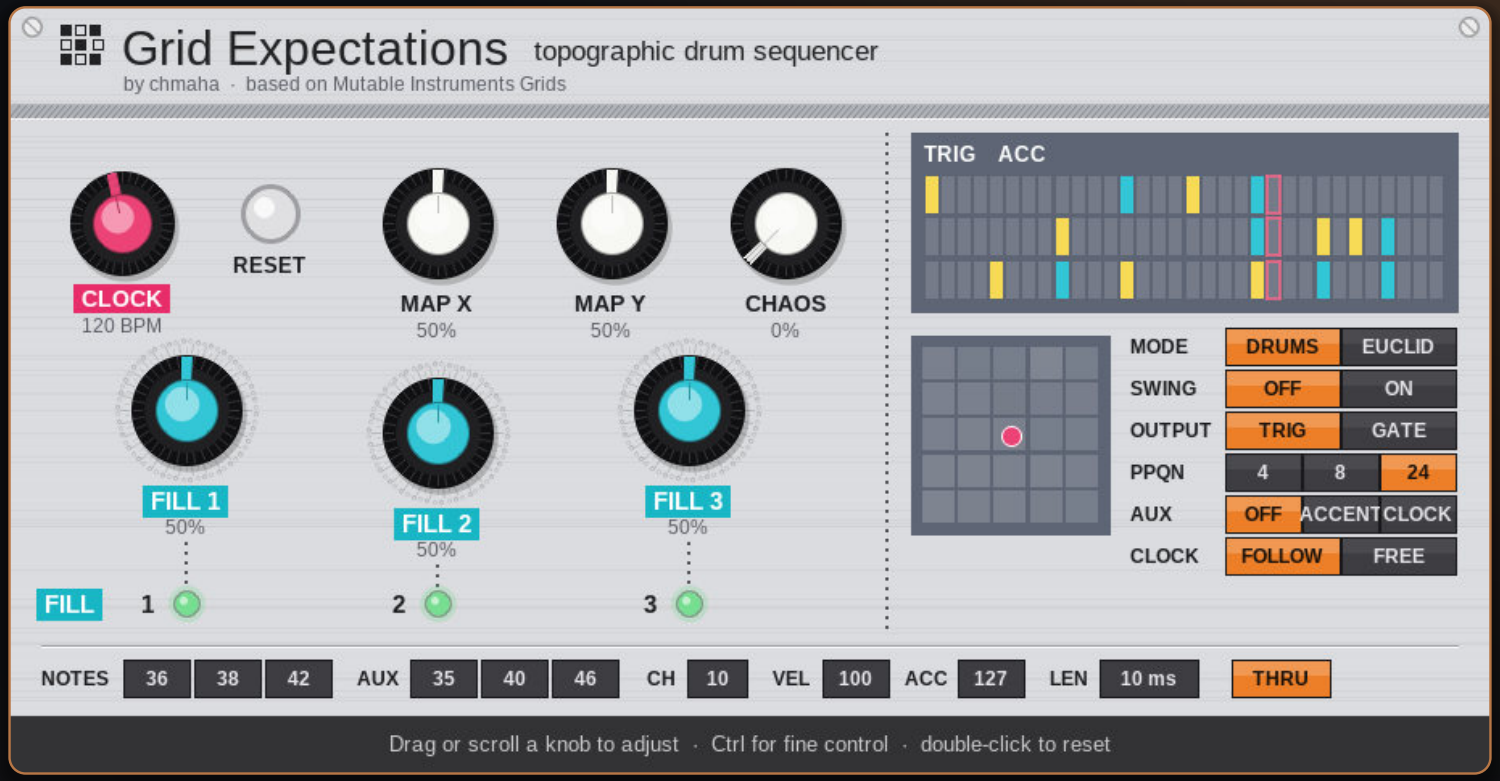
The seven noise layers, **Hum**, **Motor**, **Static**, **Noise**, **Rumble**, **Crackle** and **Crinkle**, each have an on/off switch, a **Level** and a **Pitch** (up to an octave up or down); the record on screen shows each one as it plays. **Click the record** to lift the needle and bypass the effect with a short crossfade, and click it again to drop the needle. The noise layers need the Dependencies/WaxLyrical files.


MIDI
Sequencers, generators and a patchable rack of filters. Put them ahead of an instrument.
3 ENTRIES

Grid Expectations

A topographic drum sequencer built on a 5 x 5 map of patterns.

Based on Mutable Instruments Grids



A topographic drum sequencer in the style of Mutable Instruments Grids. It makes MIDI only, so put a drum instrument after it on the same track. **MAP X** and **MAP Y** (or clicking and dragging on the map) move around a 5 x 5 map of drum patterns, blending between the nearest four. **FILL 1**, **FILL 2** and **FILL 3** set how busy the bass drum, snare and hi-hat parts are, and **CHAOS** adds random hits, rolling the dice once per bar for each part. The display at the top shows the three parts over 32 steps. Drag or scroll a knob (hold Ctrl for fine adjustment) and double-click to reset it.

MODE: **DRUMS** reads the map; **EUCLID** plays three euclidean rhythms, where MAP X, MAP Y and CHAOS set each part's length in sixteenth notes and the FILL knobs its number of hits. With **SWING** on, CHAOS sets the swing amount instead of adding hits (Drums mode only). **OUTPUT**: **TRIG** plays short notes of the set **LEN**, **GATE** holds each note for half a clock pulse. **PPQN**: 4 reads every second step, while 8 and 24 read all 32 steps and differ in gate length. **AUX** adds three extra notes: on accented hits (part resets in Euclidean mode) with **ACCENT**, or for accent, clock and bar with **CLOCK**.

CLOCK: **FOLLOW** locks to REAPER's transport, including looping; **FREE** runs at the tempo of the **CLOCK** knob, with a **RUN**/**STOP** button, and **RESET** goes back to step 1. Along the bottom, **NOTES** and **AUX** set the MIDI notes (scroll or drag them), with **CH**annel, **VEL**ocity of normal hits, **ACC**ent velocity and **THRU** to pass incoming MIDI on.



REAPER's transport or runs free with a **RUN/STOP** button. **SEED** makes the randomness repeatable. Needs the Dependencies/LosingMyMarbles files.

Losing My Marbles

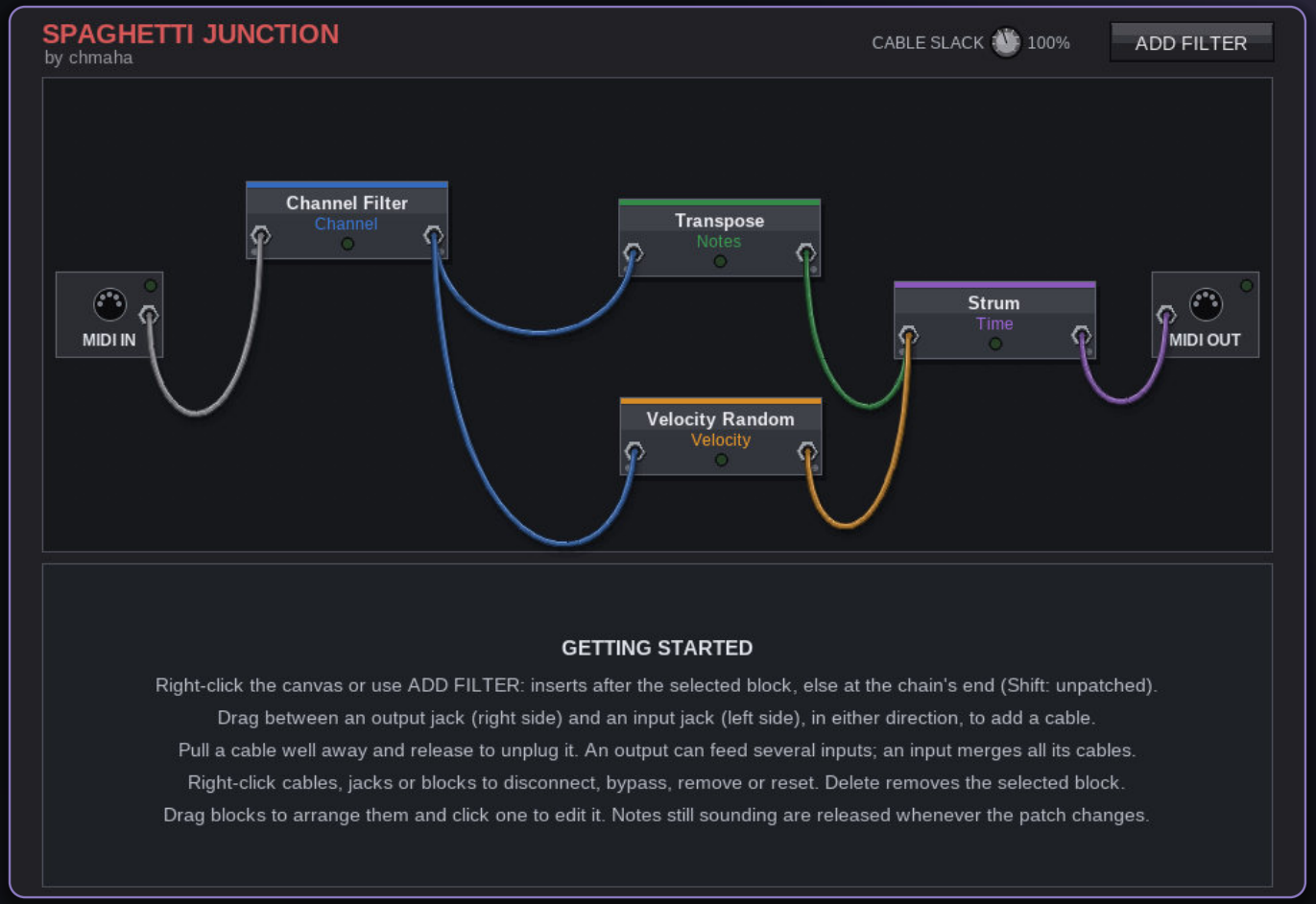
Random gates and melodies with Deja Vu looping.

Based on Mutable Instruments Marbles

A random sampler in the style of Mutable Instruments Marbles. It makes MIDI only, so put an instrument after it on the same track. Three gate streams (**t1**, **t2**, **t3**) make the rhythm and three random pitch streams (**X1**, **X2**, **X3**) make the melody; **Y** is a slower random value sent as a MIDI CC. The strip under the knobs shows them all. Drag or scroll a knob (hold Ctrl for fine adjustment) and double-click to reset it.

DEJA VU is the heart of it: to the left everything is new, at 12 o'clock it locks into a loop of the set **LENGTH**, and to the right the loop gets scrambled. The **t** and **X** buttons apply it to the gates and the pitches; click to cycle Off, On and Locked. **RATE** sets the speed (multiples of the beat from /4 to x4 under Follow), the left **BIAS** how gates divide between t1 and t3 or how sparse they are, depending on **T MODEL**, and **JITTER** humanises their timing. **SPREAD** sets how far the pitches vary, the right **BIAS** pushes them low or high, and **STEPS** glides below 12 o'clock and quantizes to the **SCALE** above it, keeping fewer notes the further it turns.

In the panel below, **T MODEL** picks how t1 and t3 share the clock (coin toss, clusters, drums, independent, dividers, three states or Markov). **VOICE**: **Pitch** plays each gate's X value through the scale from **ROOT**; **Drums** plays a fixed note per gate. **X CTRL** sets how the X controls spread across the three streams, **X CLOCK** which gates clock them, and **X INPUT** set to **MIDI In** shifts incoming notes through the Deja Vu register. **CLOCK** follows



Spaghetti Junction

Patch the 34 x42 MIDI filters together with cables.

Based on x42 midifilter.lv2

A rack for the 34 x42 MIDI filters. Build a patch by connecting **MIDI IN** to **MIDI OUT** through up to eight filter blocks.

- **Add a filter**: right-click the canvas or use **ADD FILTER**. It goes after the selected block, or at the end of the chain if nothing is selected (hold Shift to leave it unpatched).
- **Connect**: drag between an output jack (right side) and an input jack (left side), in either direction.
- **Disconnect**: pull a cable well away from its path and release.
- **Edit**: drag blocks to arrange them and click one to edit it. Right-click cables, jacks or blocks to disconnect, bypass, remove or reset. Delete removes the selected block.
- One output can feed several inputs, and an input merges every cable plugged into it, so you can build chains, splits and layers.
- **CABLE SLACK** sets how loosely the cables hang. Held notes are released whenever the patch changes.

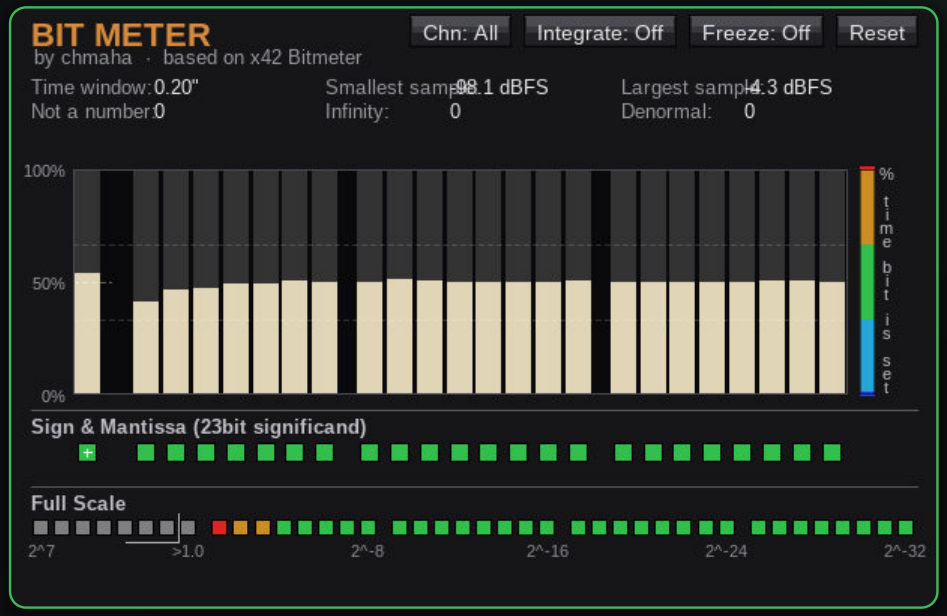
Requires REAPER 7 or later.


METERS
None of these change the sound.
14 ENTRIES

Bit Meter

See which bits are really in use.

Based on x42 Bitmeter



Shows which bits are in use. **Freeze** holds the display, **Integrate** averages over time, **Chn** picks All, L or R, and **Reset** clears it. It also reports the smallest and largest sample and flags denormals, infinities and NaNs.

K-Meter

K-12, K-14, K-20 or anything between.

Based on x42 K-Meter

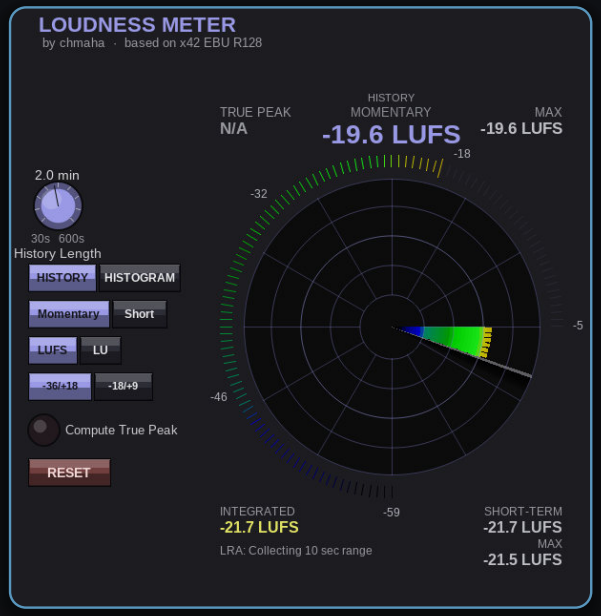


Pick the **K-12**, **K-14** or **K-20** scale (or any K value between 12 and 23). Peak and RMS are shown, and the **dBFS** button switches the readout to dBFS.

Loudness Meter

EBU R128 loudness on a radar display.

Based on x42 EBU R128

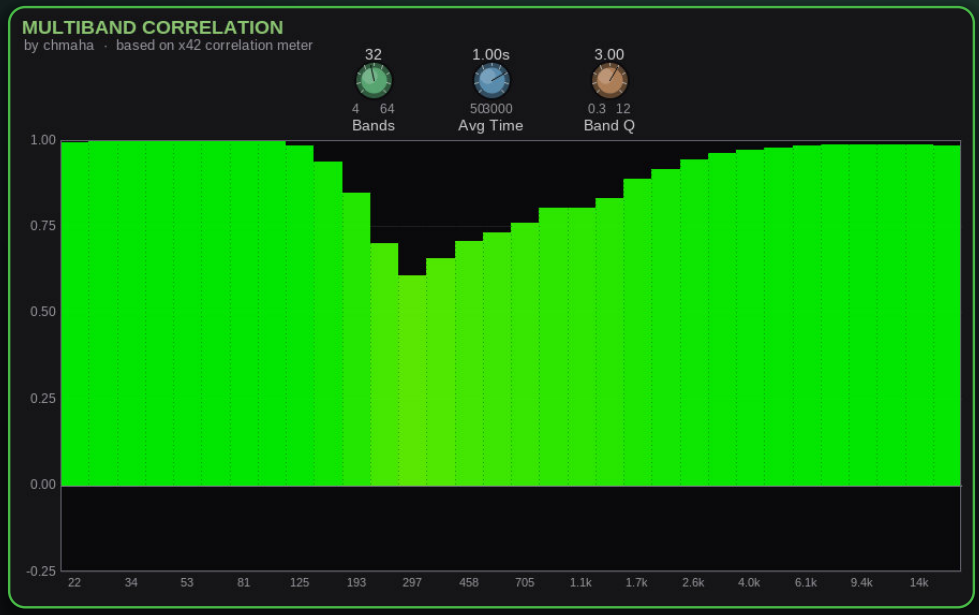


A radar display plus numeric readouts for momentary, short-term and **INTEGRATED** loudness, **RANGE (LRA)**, **MAX** and **TRUE PEAK**. Choose **HISTORY** or **HISTOGRAM** for the radar, **Momentary** or **Short** as its source, the **History Length**, and **LUFS** or **LU** units. **RESET** clears the measurement. True peak is only computed if you switch **Compute True Peak** on.

Multiband Correlation Meter

Stereo correlation, band by band.

Based on x42 correlation

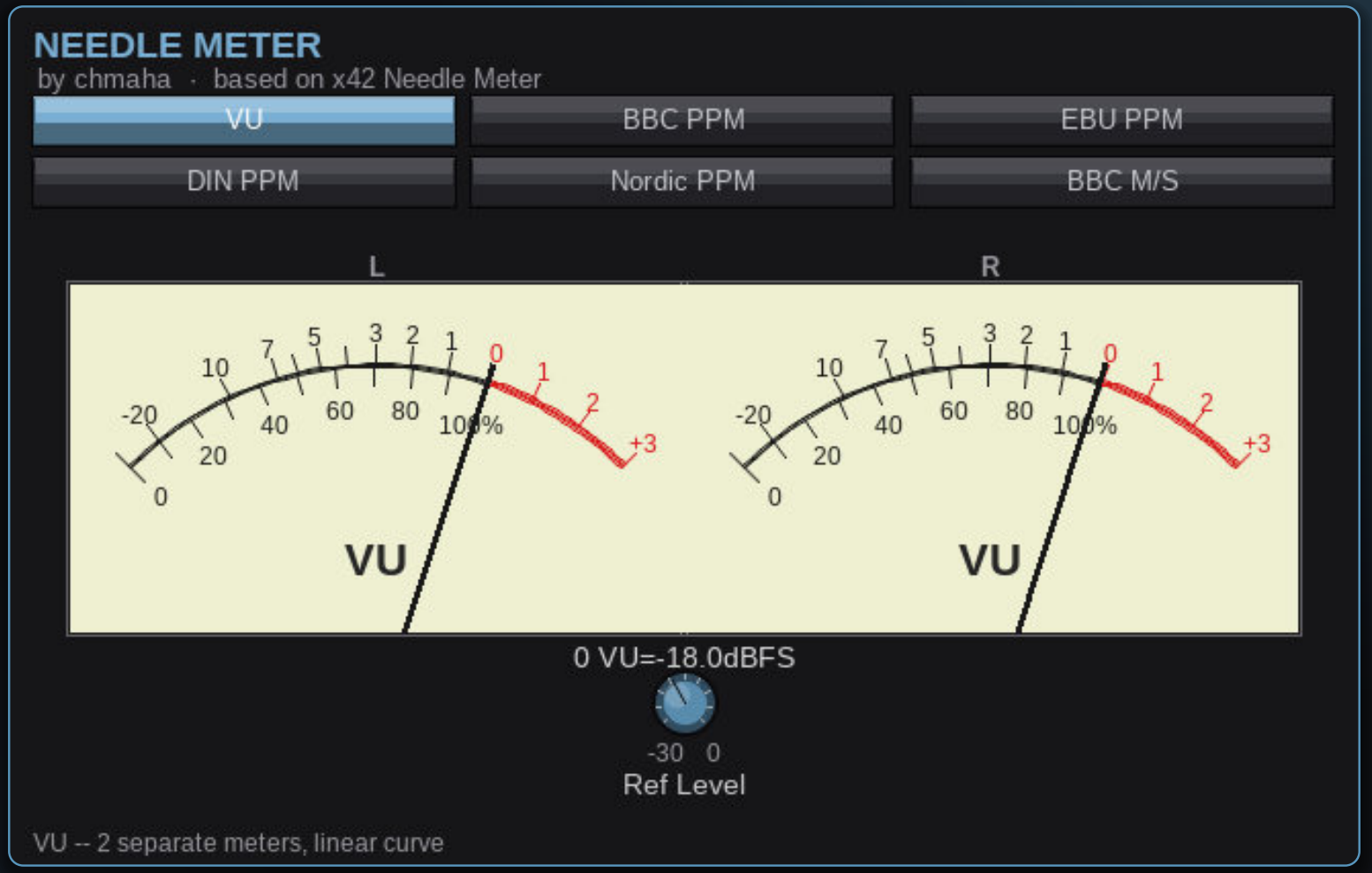


Stereo correlation per frequency band. Knobs: **Bands**, **Avg Time** and **Band Q**.

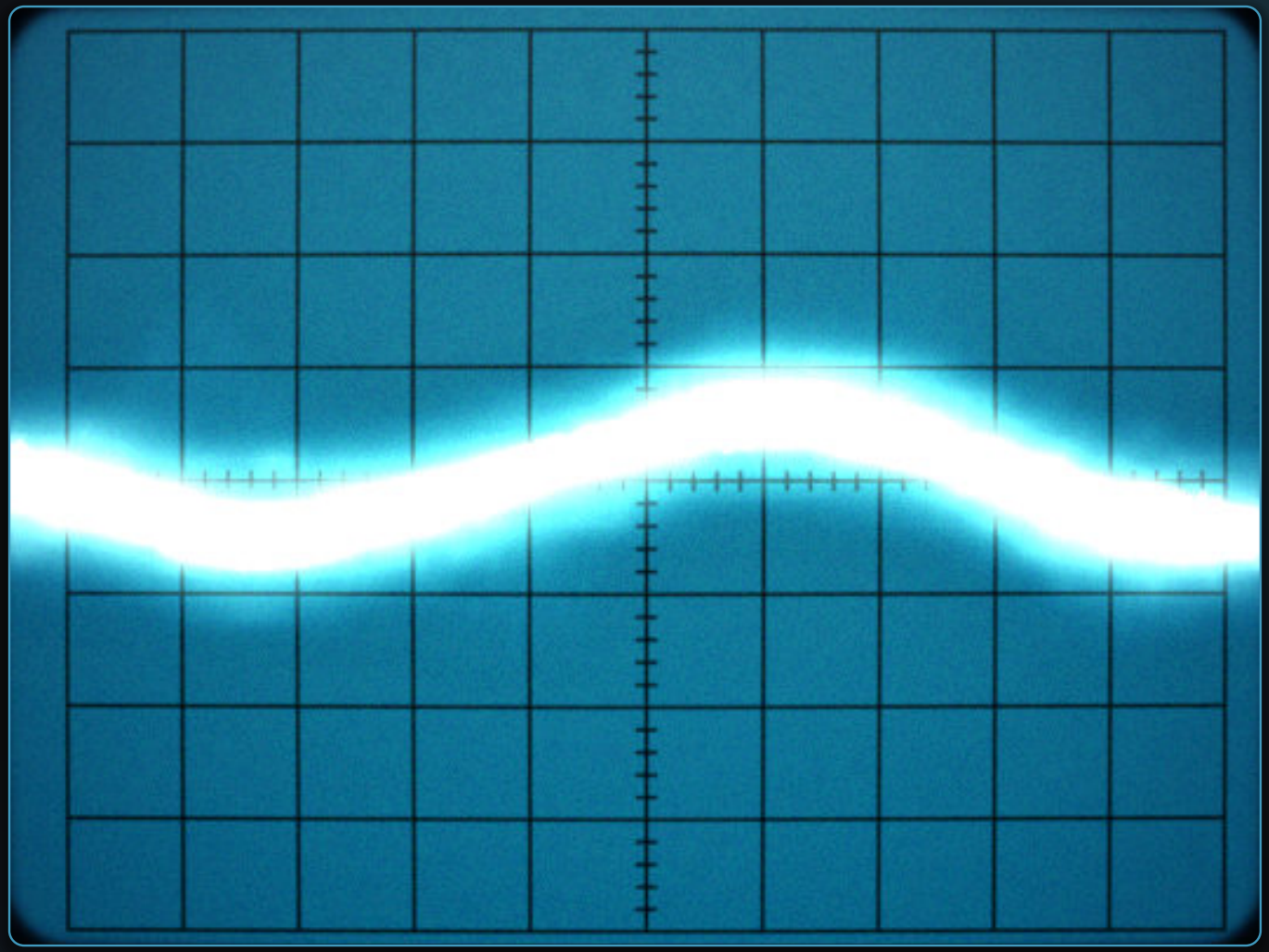
Needle Meter

VU and five flavours of PPM.

Based on x42 Needle Meter



Choose a style: **VU**, **DIN PPM**, **EBU PPM**, **BBC PPM**, **Nordic PPM** or **BBC M/S**. **Ref Level** sets which dBFS level is the reference, and the **TEST** mark shows the alignment level.



Oscilloscope

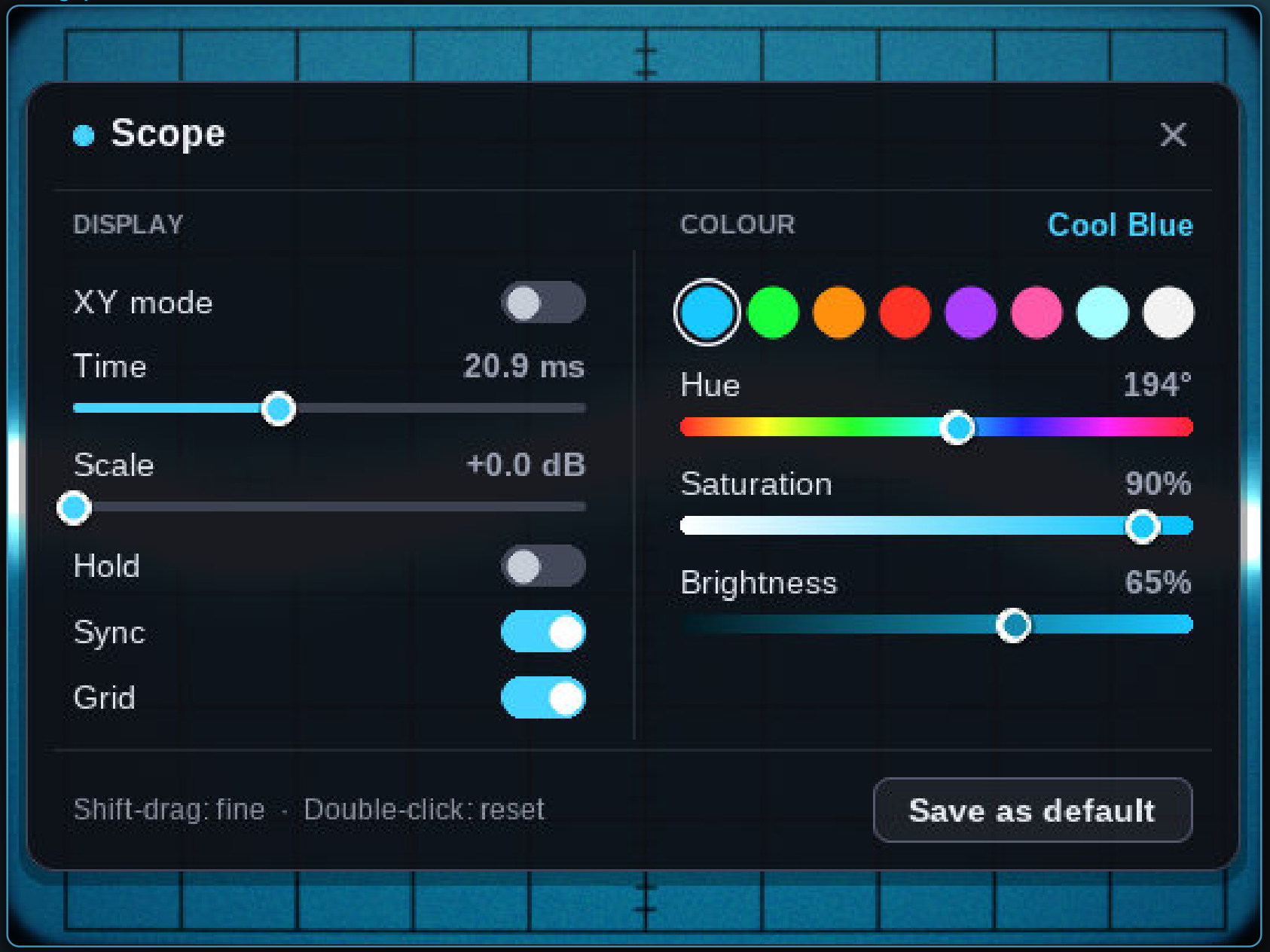
A scope with XY mode, sync and your choice of colour.

Based on unplugred's Scope

Click the scope to open or close its settings panel. Settings: **XY** mode, **Time**, **Scale**, **Hold**, **Sync**, **Grid** and the colour (hue, saturation, value). Shift-drag for fine adjustment, double-click to reset.

Oscilloscope

Settings panel

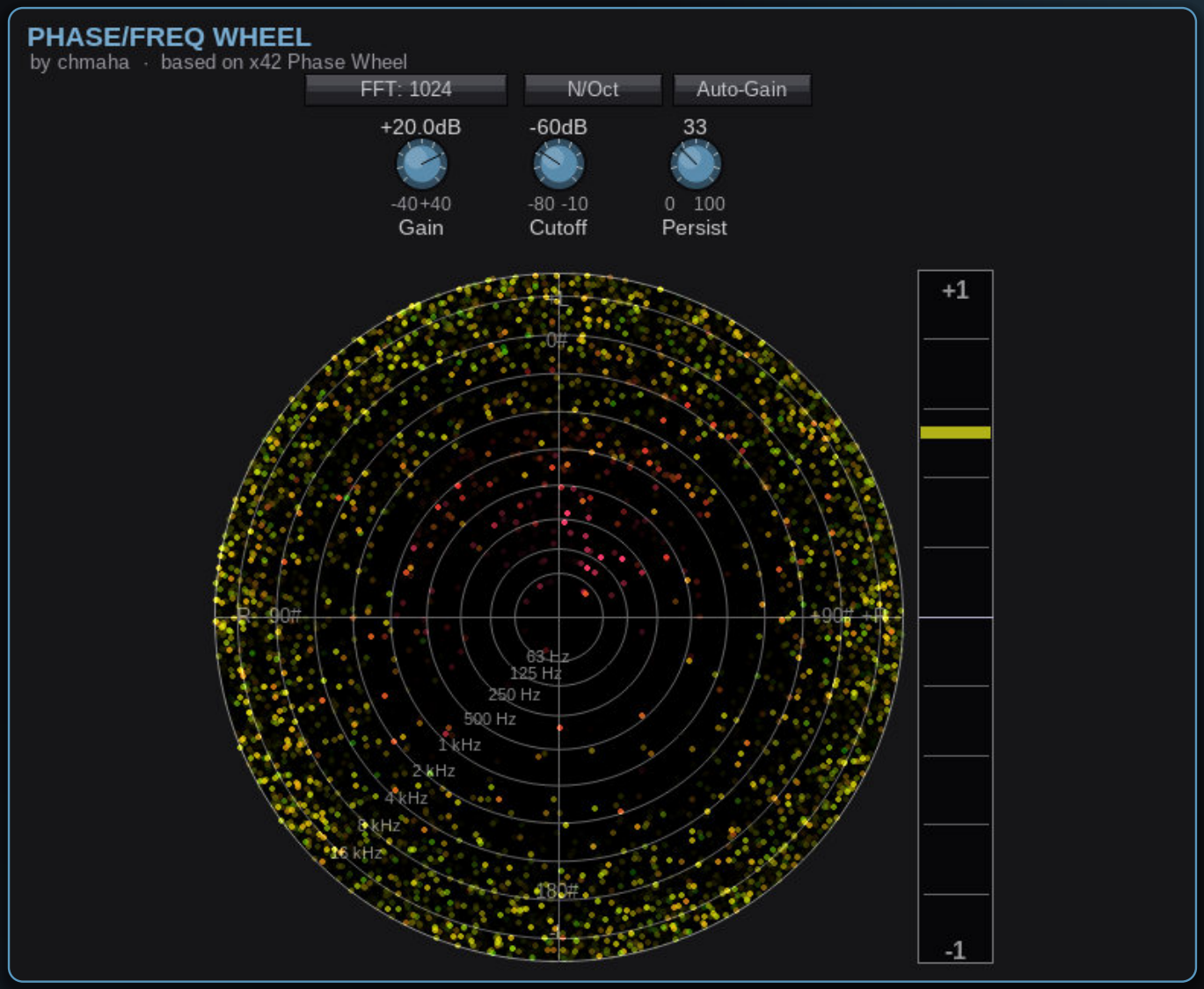


Phase/Frequency Wheel

Stereo phase against frequency.

Based on x42 Phase Wheel

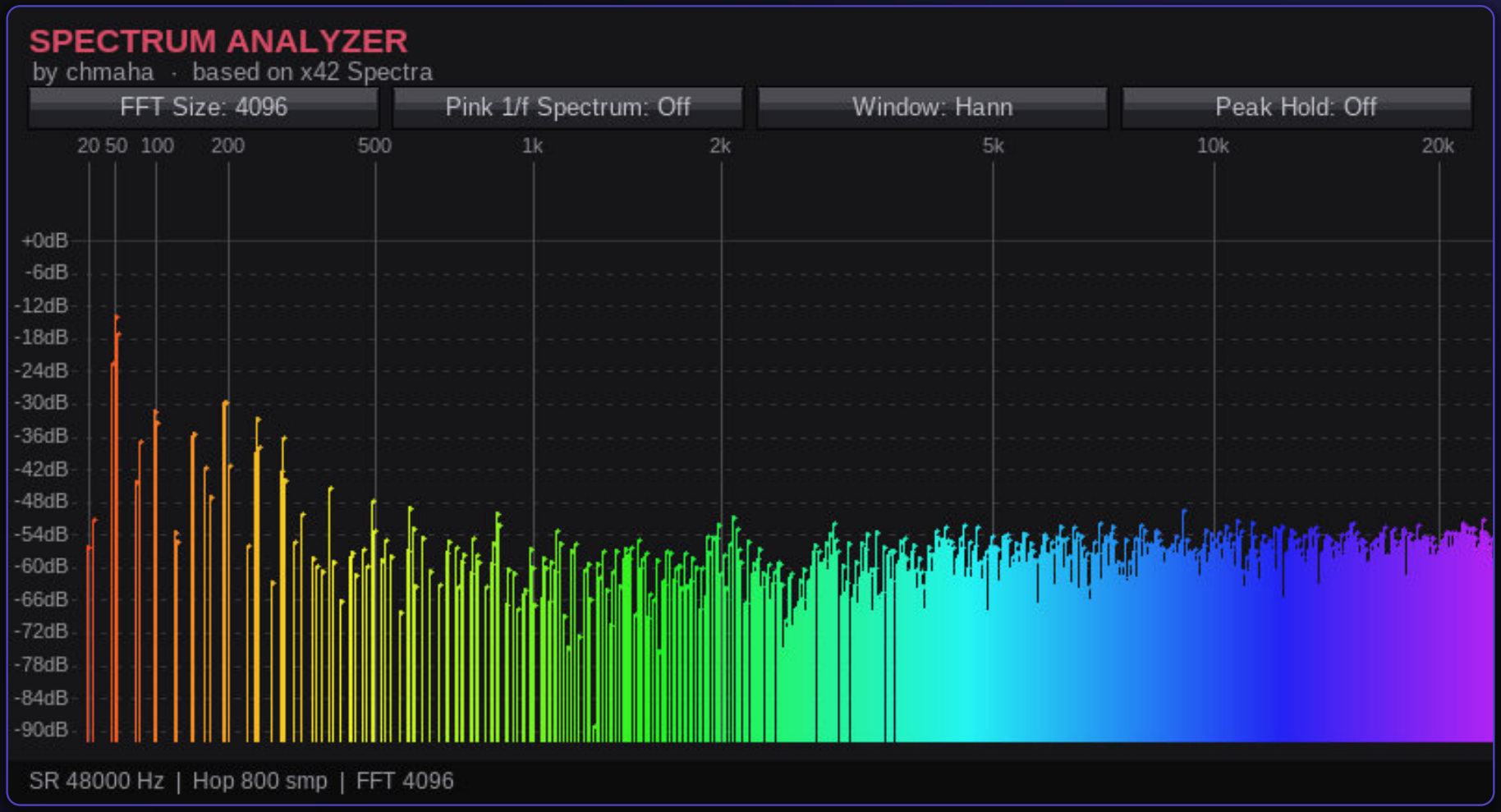
Shows stereo phase against frequency. Knobs: **FFT** size, **N/Oct** (bands per octave), **Persist**, **Gain** (or **Auto-Gain**) and **Cutoff**.



Spectrum Analyzer

An FFT spectrum with peak hold and a pink-noise view.

Based on x42 Spectra



Buttons: **FFT Size**, **Window** (Hann, Blackman-Harris or Flat Top), **Pink 1/f Spectrum** (flattens pink noise) and **Peak Hold**.

Stereo Phase Scope



A goniometer.

Based on x42 Goniometer

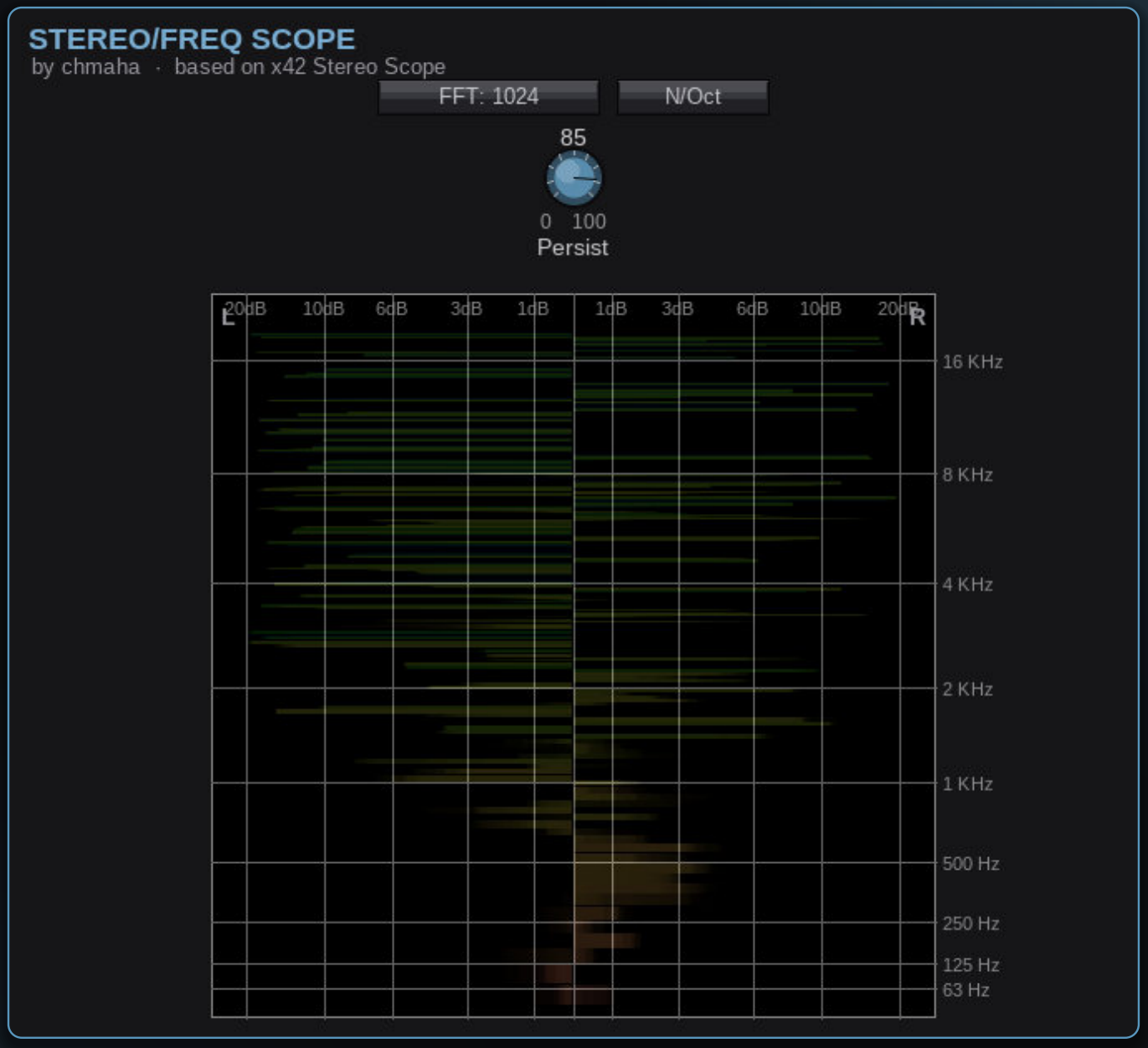
A goniometer. Open **Settings** for **Gain** (with **Autogain**), **Attack**, **Decay**, **Target** zoom, **RMS/Peak**, **Lines** or points with **Line Width** and **Point Size**, **Oversample**, **Persistence** and **Limit FPS**.

Stereo/Frequency Scope

Stereo image by frequency.

Based on x42 Stereo Scope

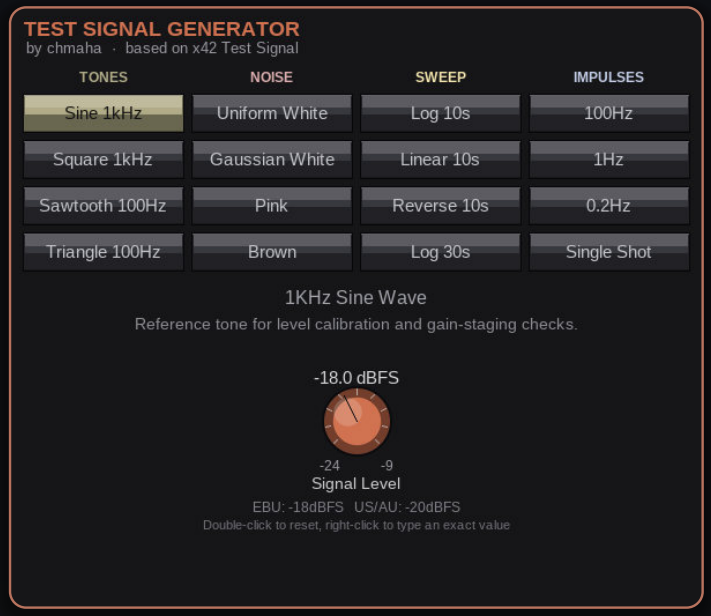
Stereo image by frequency. Knobs: **FFT** size, **N/Oct** and **Persist**.



Test Signal Generator

Tones, noise, sweeps and impulses.

Based on x42 Test Signal

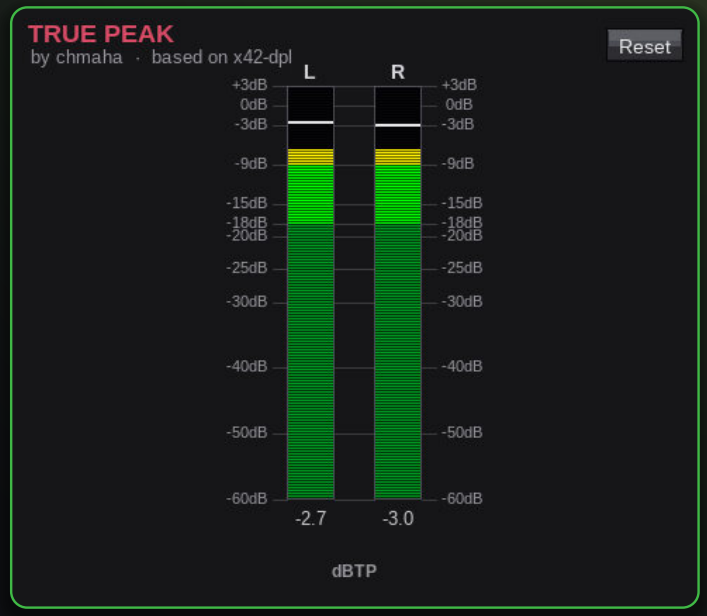


Pick a signal: 1 kHz sine or square, uniform or Gaussian white noise, pink or brown noise, 100 Hz impulses, sawtooth or triangle, a single impulse, slow impulses (1 Hz and 0.2 Hz), and log or linear sine sweeps. Set the **Signal Level** (EBU reference is -18 dBFS and US/AU is -20 dBFS). Double-click to reset it, or right-click to type an exact value. Lower your monitoring before you start.

True Peak

A true-peak (dBTP) meter.

Based on x42 dpl

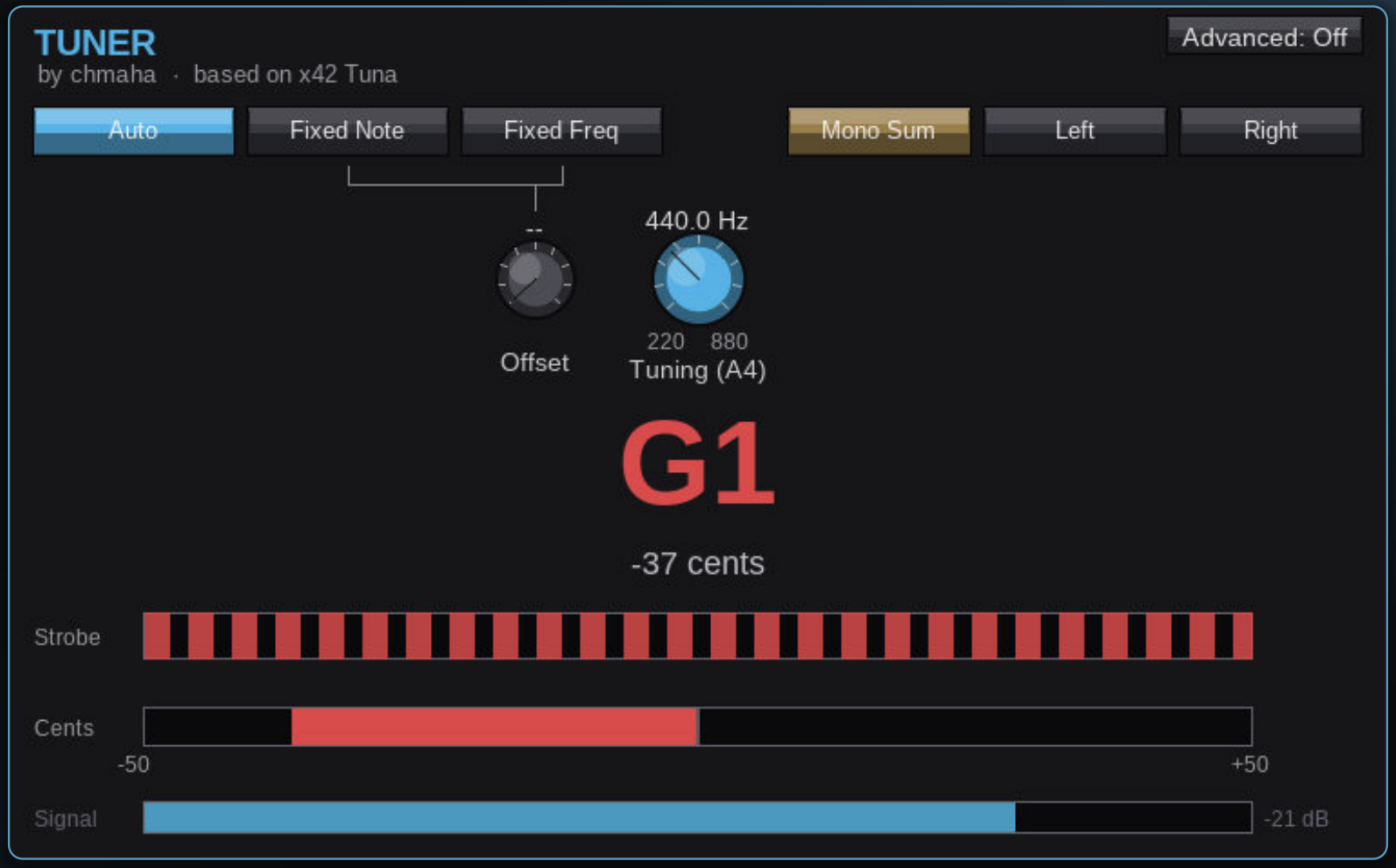


A true-peak (dBTP) meter. **Reset** clears the held peak, and a warning appears when the signal goes over 0 dBTP.

Tuner

A strobe tuner with adjustable detection.

Based on x42 Tuna



Choose **Auto**, **Fixed Note** or **Fixed Freq**, and set the reference pitch under **Tuning (A4)**. A strobe display shows the note and cents offset. **Advanced** reveals the detection thresholds (**RMS**, **PostFilt**, **FFT**, **Override**, **Fundmntl**, **Octave**, **Overtone**) and the input source (**Mono Sum**, **Left** or **Right**).

VU Meter

A VU needle with adjustable nominal level and speed.

Based on unplugred's VU



Click the meter to open or close its settings panel. Settings: **Nominal** level, **Rise** and **Decay** speed, and **Mono** or **Stereo**. Shift-drag for fine adjustment, double-click to reset.

VU Meter

Settings panel





VISUALS

Something to look at while it plays.

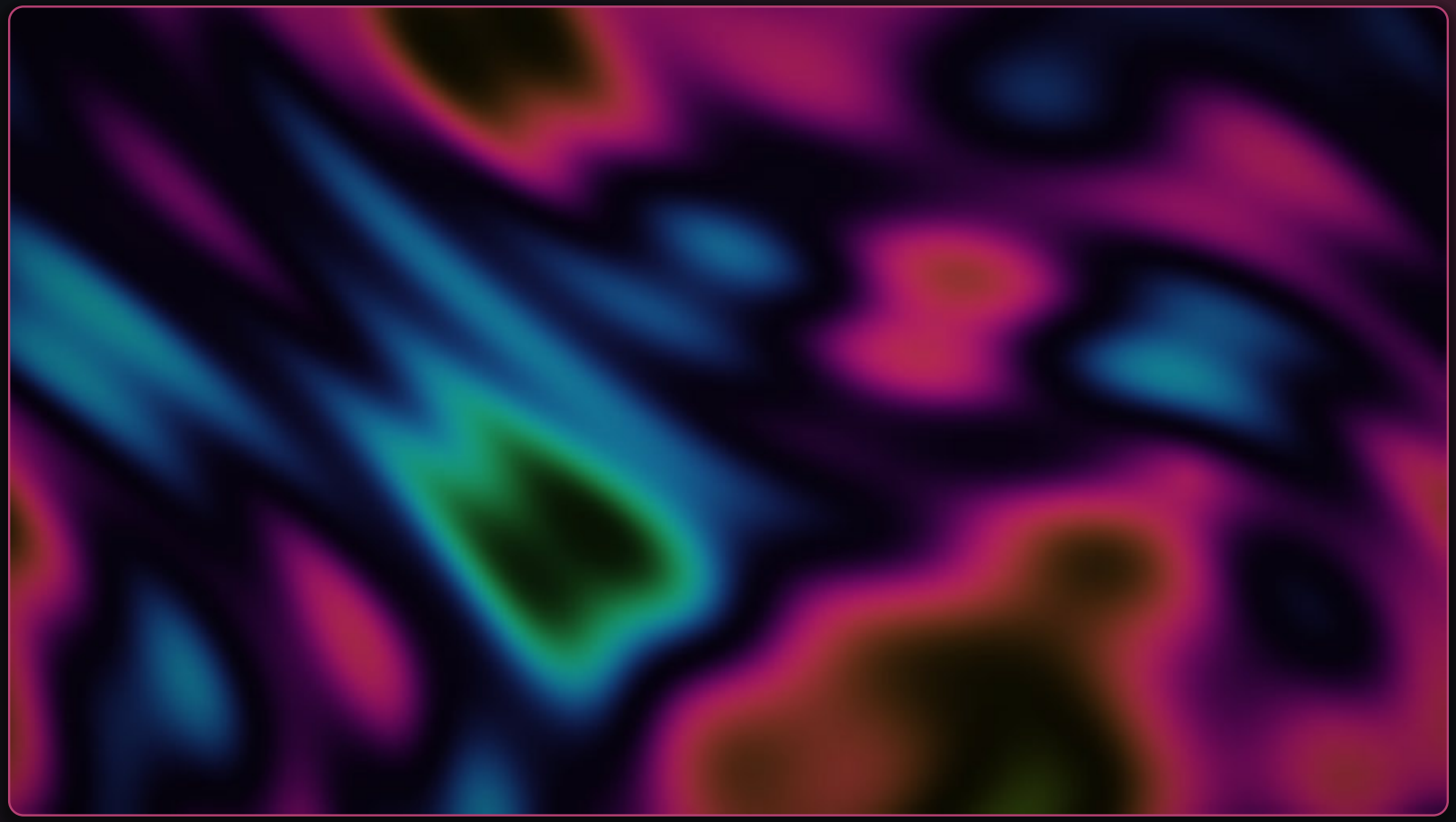
1 ENTRY

VISUALIZATION

Sight for Sore Ears

An audio-reactive visualizer with 25 scenes.

Original design



An audio-reactive visualizer; the sound passes through untouched. **Click the picture** to open the settings panel, and click the **×** or anywhere outside the panel to close it. Pick one of the 25 scenes from the grid, from **Plasma** to **Harmonograph**, and one of the 12 colour palettes from the swatches below. **Quality** (**Low**, **Med**, **High**, **Ultra**) trades detail for CPU. Drag **Sensitivity** to set how strongly the picture reacts and **Speed** to set how fast it moves; double-click either to reset it to 50%. In **Live Score**, **Live Score look** chooses **Smoke** or **Paint**. The mouse wheel steps through the choice under the mouse.

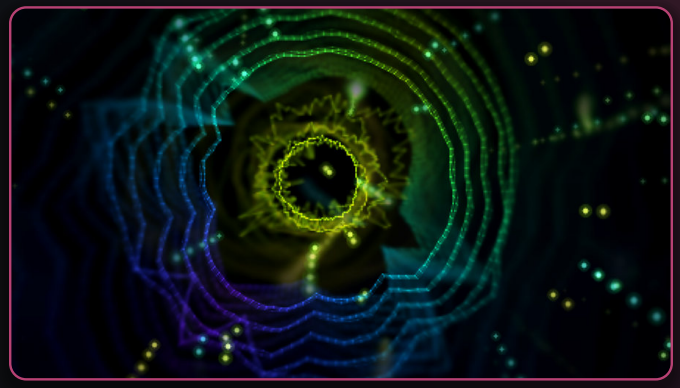
The scenes follow drums on the hit; in music without drums (classical, ambient), note onsets take over, and loudness is judged against the recent level so quiet passages stay alive.

Sight for Sore Ears: the scenes (1 of 3)

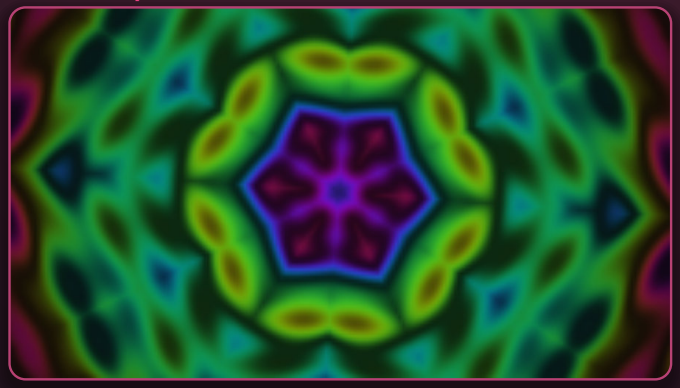
Lava Lamp



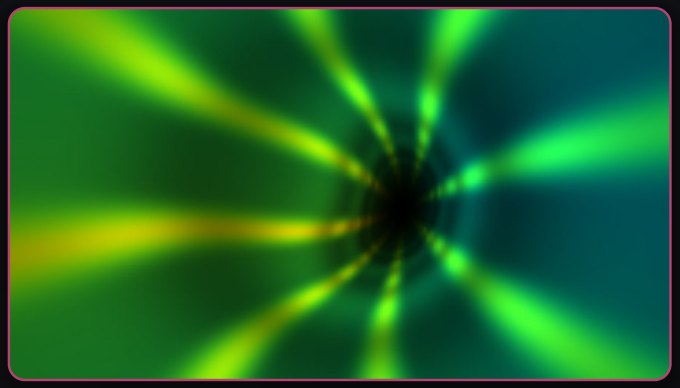
Nebula



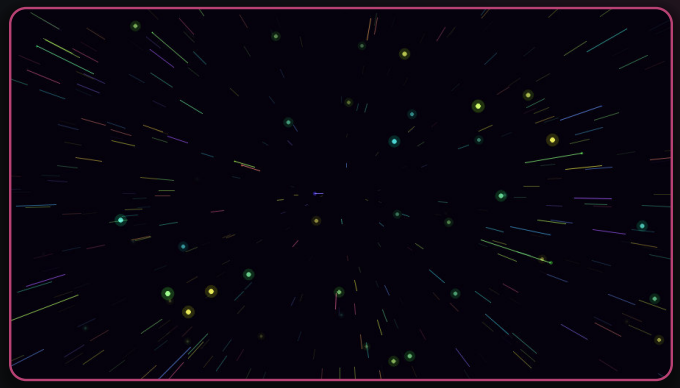
Kaleidoscope



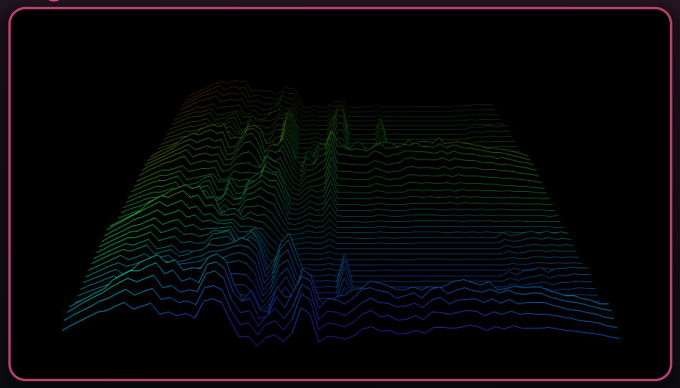
Tunnel



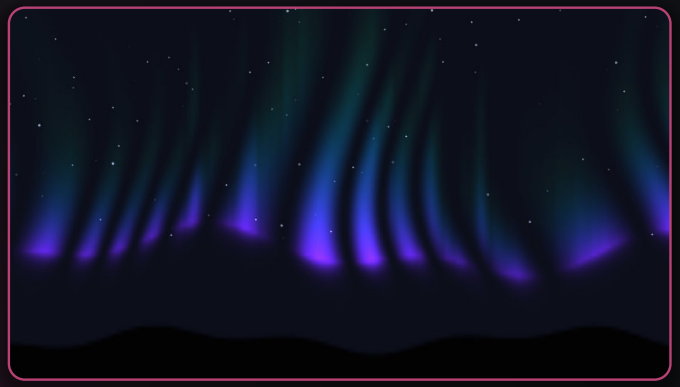
Starfield



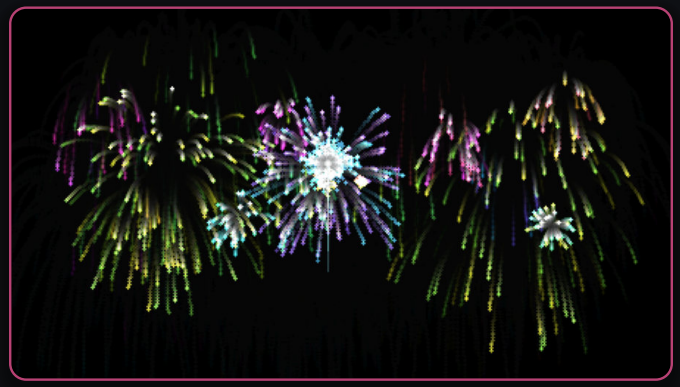
Ridgelines



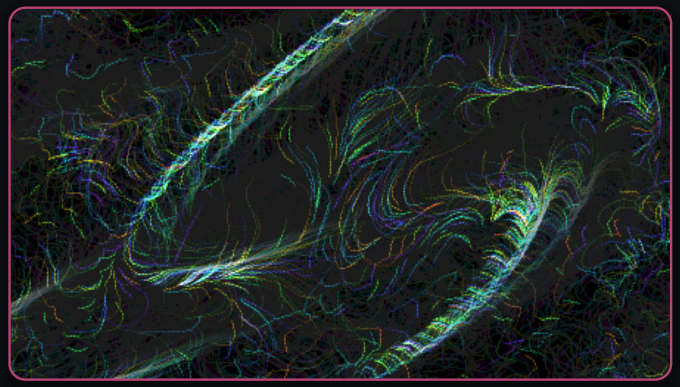
Aurora



Fireworks

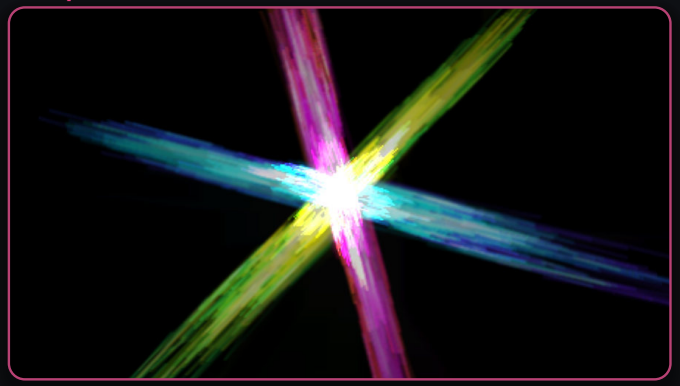


Silk



Sight for Sore Ears: the scenes (2 of 3)

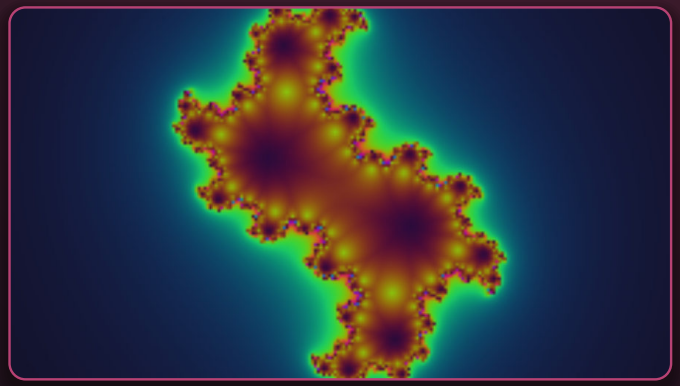
Phosphor



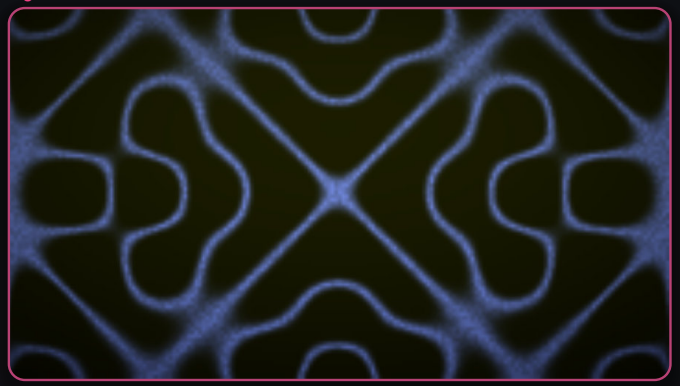
Stained Glass



Julia



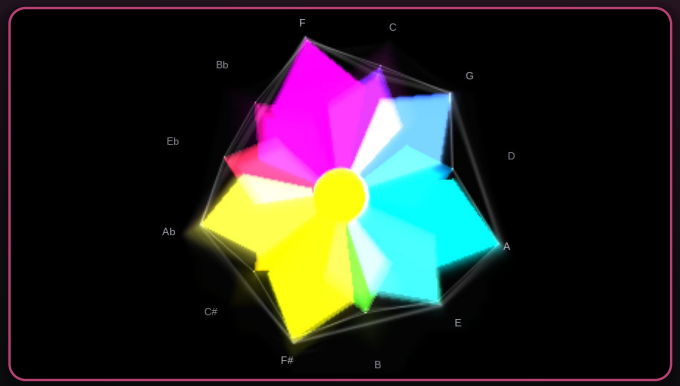
Cymatics



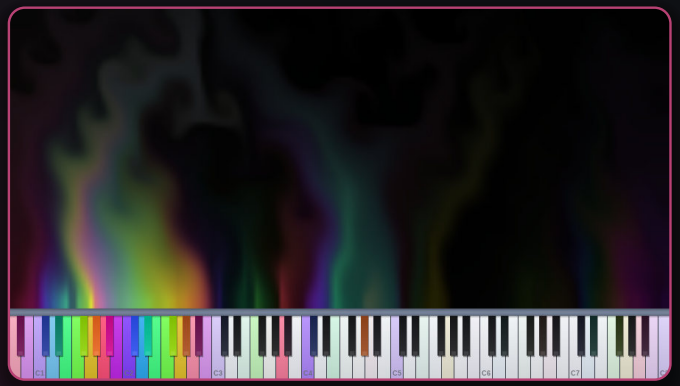
Skyline



Chroma Wheel



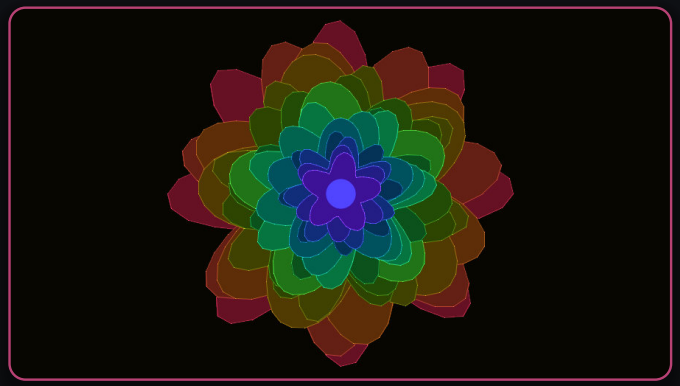
Live Score



Constellations



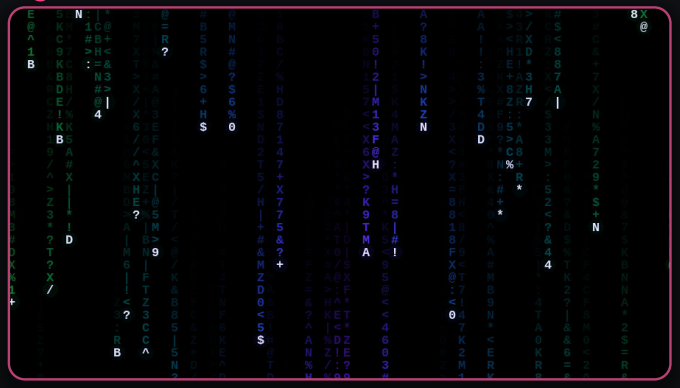
Bloom



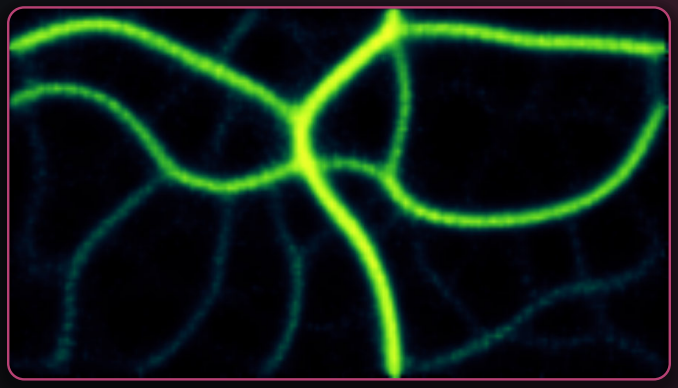
VISUALIZATION

Sight for Sore Ears: the scenes (3 of 3)

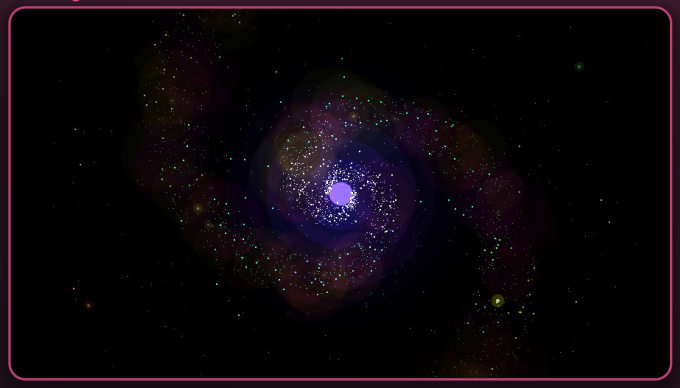
Digital Rain



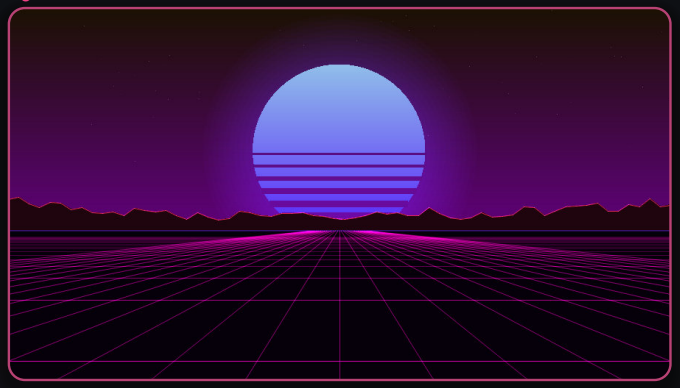
Slime



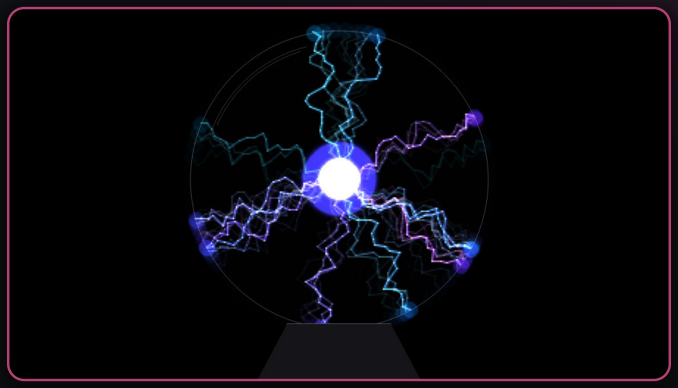
Galaxy



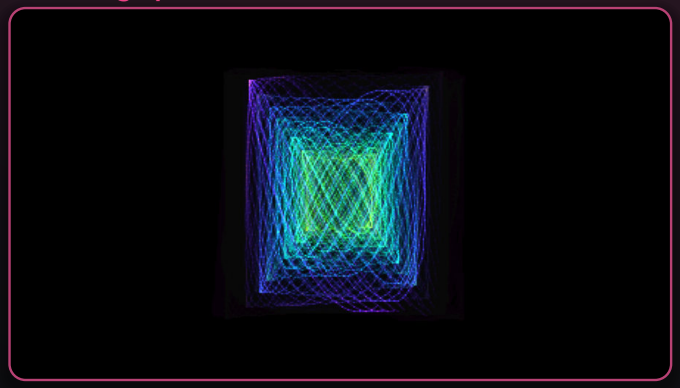
Synthwave



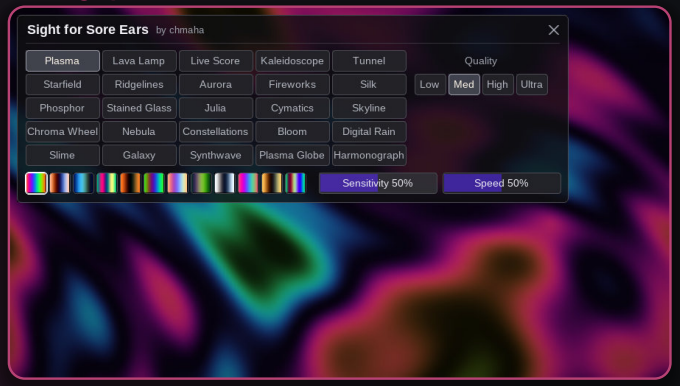
Plasma Globe



Harmonograph



Settings panel





UTILITY

Small jobs, done properly.

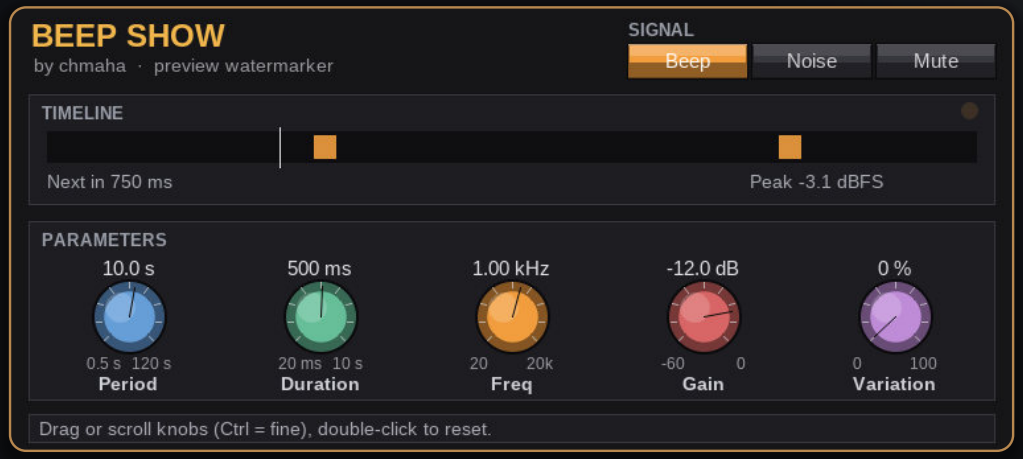
3 ENTRIES

UTILITY / 1 OF 3

Beep Show

Watermark demos, previews and stems with beeps, noise or dips.

Original design, after the feature set of Voxengo Beeper



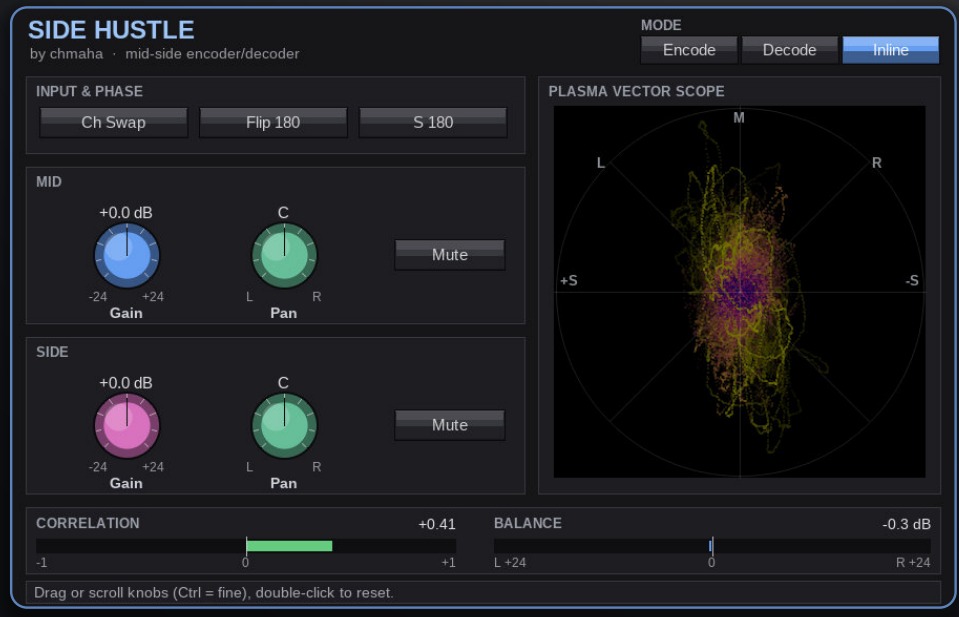
A preview watermarker. **Signal** chooses what is put into the audio: **Beep** (a sine tone), **Noise** (a low-passed noise burst, harder to remove cleanly) or **Mute** (a dip in level). **Period** sets the time between insertions, **Duration** how long each one lasts, and **Freq** the beep pitch or the noise filter's corner frequency. **Gain** sets the beep or noise level in dBFS; the material dips in proportion, so the output never goes above its highest peak since playback started (a beep louder than that peak mutes the material for its length). With Mute, the knob becomes **Mute To**, the level the material drops to (-60 dB mutes it completely). **Variation** randomises the timing, duration, frequency and gain of every insertion. The timeline shows when the next insertion comes and the peak so far. Drag or scroll a knob (hold Ctrl for fine adjustment) and double-click to reset it. The pattern follows the project position, so every render is identical.

UTILITY / 2 OF 3

Side Hustle

Encode, decode and adjust mid and side.

Original design, after the feature set of Voxengo MSED



A mid-side encoder, decoder and processor. **Mode** picks the job: **Encode** turns left/right into mid (left output) and side (right output), **Decode** turns mid/side back into left/right, and **Inline** encodes, adjusts the mid and side channels and decodes again, all in one instance. **Ch Swap** swaps the input channels, **Flip 180** inverts the polarity of both channels and **S 180** inverts the side channel only.

Mid and Side each have **Gain** (±24 dB), **Mute** and **Pan** (0 dB cosine pan law; pan works in Decode and Inline). Right-drag either Gain knob to move both in opposite directions. The **correlation** meter (500 ms average) reads -1..0 for anti-phase, 0 for unrelated and 0..1 for in-phase stereo; the **balance** meter shows how much louder one side is (negative means left is louder). The plasma vector scope shows the output with each octave band at its own radius, lows on the outside and highs towards the centre: centred signals draw a vertical line, panned signals a diagonal, out-of-phase signals a horizontal line and unrelated signals a circle. Zero latency.



Stereo Typical

Gain, polarity, pan, width and rotation, with a goniometer.

Original design, after the controls of FLUX:: Stereo Tool

A stereo imaging tool. **Left** and **Right** each have **Input Gain** (±12 dB), **Phase** (polarity invert), **Mute** and a pan pot (constant-power pan law; hard left and hard right by default). **Width** to the right widens the stereo image, to the left narrows it down to mono. **Rotation** turns the whole image up to 45° towards the left or right without losing level.

The **preset** menu at the top applies one of ten presets straight away (Default, Mono, Swap L/R, Left to Both, Right to Both, Wide, Narrow, Left Phase, Right Phase and Half Width Pans) and shows which one matches the current settings, or Custom. Input and output peak meters sit either side; click a peak readout to reset it. The phase correlation meter and the goniometer show the output: centred mono is vertical, left or right panned mono lies on the L or R diagonal, out-of-phase signals turn horizontal and very wide signals spread sideways. **Normalize** scales the goniometer so the recent peak fills the display, which helps with quiet material.

CREDITS

Standing on shoulders.

Every plugin here exists because someone wrote and shared the original. Each plugin page names its source, and the header of each .jsfx file gives the details.

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- **GPLv3 or later** (JSFX code): ports of GPL originals, including the x42 plugins (GPLv2 or later), zam-plugins (GPLv2 or later), Zita-Rev1, Aeolus, setBfree, amsynth, Dexed, mda-lv2, mda-effects (Lip Service, MIT or GPLv2 or later), Spectral Compressor, CHOW Tape Model and ZynAddSubFX, plus Acid Reign (from the MIT-licensed Open303), Wax Lyrical (from Calf Vinyl and FluidSynth, both LGPL 2.1), Console Yourself (from Nes_Snd_Emu and snes_spc, both LGPL 2.1 or later, and the BSD-3-Clause MAME sound cores and ymfm), Skin in the Game (from the GPLv3-or-later 4ms CoreModules), Grid Expectations (from the GPLv3 Grids) and the other Mutable Instruments ports: Ring Leader, Silver Lining, Plait Tectonics, Bead Time Story, In Your Element, Warp Factor and Losing My Marbles (from the MIT-licensed eurorack code, including the tables in Dependencies).
- **MIT**: JuKnow Chorus (Peter Allwin), ChronoDupe (Christopher Arndt), Analog EQ and Final Touch (lkjb), and the Airwindows ports (Chris Johnson).
- **AGPLv3 or later**: the unplugred ports (Oscilloscope, VU Meter, Super Color). unplugred's image assets and samples are CC BY-NC (non-commercial).
- **Samples**: the AVL Drumkits samples and photos by Glen MacArthur are CC-BY-SA 3.0, and music made with the kits can be licensed freely.
- **SCC Expressive Strings** (Espressivo String Ensemble): released by S. Christian Collins as "use at your own risk", with samples collected from free SoundFont banks.
- **Console Yourself SNES samples**: Doug Fraker's BRR samples are public domain per their author (repository MIT).
- **jRhodes3d samples** (Tine Capsule): CC BY-NC 4.0 by Jeff Learman (non-commercial).
- **Greg Sullivan E-Pianos samples** (Grand Designs, Sticky Situation, Out of This Wurld): CC BY 4.0 by Greg Sullivan; SFZ mapping by kinwie, CC BY 3.0.
- **Sonivox EAS** (Major General MIDI): Apache 2.0 original.

The GPLv3 text is in LICENSE. The MIT notices are included in the JuKnow Chorus and ChronoDupe sources.



THE JSFX VERSIONS

JSFX ports, interfaces and original designs by **chmaha**.

Claude was used in the creation of this product.

github.com/chmaha/DSP-Ports-JSFX

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